

CLM Planning, Installation, and Service

LK02-100

Communication Link Module (CLM)

CLM ***Planning, Installation, and Service***

LK02-100
01/09

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About This Publication

This publication provides the general description, site preparation, installation, check out and service instructions for the Communication Link Module (CLM) to an AM with CLM communication board(s) installed.

**For installation to a TPS/CLM (APP-based) refer to the
*TPS/CLM Planning and Installation Manual.***

Appendix A gives further information about cable connections to be used in connecting the Communication Link Module Interface (CLMI) to Bristol-Babcock RTUs, including modem connections.

Appendix B gives further information about cable connections to be used in connecting the CLMI or CCP boards to Modbus compatible device networks.

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World Wide Web	Honeywell Solution Support Online: http://www.honeywell.com/ps		
Training Classes	Honeywell Automation College: http://www.automationcollege.com		

Table of Contents

1. INTRODUCTION	11
1.1. GENERAL DESCRIPTION	11
1.2. HONEYWELL SUPPORT SERVICES.....	12
1.3. RELATED PUBLICATIONS	12
2. SITE PREPARATION.....	13
2.1. STORAGE CONDITIONS.....	13
2.2. SITE REQUIREMENTS	13
2.2.1. Dimensions and Weight	13
2.3. ELECTRICAL REQUIREMENTS.....	14
2.3.1. AC Voltage Options	14
2.3.2. Frequency	14
2.3.3. Current in Amperes at 120 Vac	14
2.4. CONFIGURATIONS (CLMI/CCP).....	15
2.4.1. Cabling Considerations	15
2.5. CONNECTIONS (CLMI/CCP)	16
2.5.1. Modem Usage and Selection	16
2.5.2. Non-redundant CLM Port Connections.....	17
2.5.3. Redundant CLM Port Connections	18
3. HARDWARE INSTALLATION	19
3.1. UNPACKING	19
3.2. ASSEMBLY AND CABLING.....	19
3.2.1. Module Installation.....	19
3.2.2. LCN Node Pinning.....	20
3.2.3. CLMI, CCP, UCIM Pinning.....	22
3.2.4. CLMI/CCP 4-Port Interface Board	22
3.2.5. Cabling and Pinning for Modems.....	23
3.2.6. Cabling for RS485 Networks	24
3.3. INSTALLATION WRAP-UP	26
4. CHECKOUT	27
4.1. POWER-ON TESTS	27
4.1.1. Electronics Checks	27
5. CLM PLATFORM SOFTWARE INSTALLATION	29
5.1. Introduction	29
5.1.1. AM-based Installation	29
5.1.2. Configuration overview.....	29
5.2. Install CLM Platform Hardware - Summary.....	30
5.2.1. Procedure.....	30
5.2.2. Hardware Installation	30
5.3. Install CLM Platform Software	31

5.3.1.	Before You Do the Software Installation	31
5.3.2.	Execute the Install Procedure	32
5.3.3.	Build Required CLM Platform Points	33
5.3.4.	Customize Displays	35
5.3.5.	Configure the CLM Interface Board (CLMI,CCP,UCIM).....	37
5.4.	Update LCN Network Configuration File	39
5.4.1.	NCF update process	39
5.4.2.	CLM is External Load Module (.LO)	39
5.4.3.	Modify NCF	41
5.5.	Load the AM with CLM Platform Software.....	43
5.5.1.	Restart the node.....	43
5.6.	Add Other CLMs After the First Installation	44
5.6.1.	Adding other CLMs	44
5.6.2.	Summary procedure	44
6.	SERVICE	45
6.1.	OVERVIEW	45
6.2.	FIELD ADJUSTMENT	47
6.3.	GENERAL TROUBLESHOOTING	47
6.4.	CLMI and CCP TROUBLESHOOTING	48
6.4.1.	CLMI Hardware Indicators	48
6.4.2.	CLMI Hardware Indicator Troubleshooting Chart	50
6.4.3.	CLMI System Software Indications	50
6.4.4.	CCP Hardware Indicators.....	51
6.4.5.	CCP Hardware Indicators.....	52
6.4.6.	CCP Hardware Indicator Troubleshooting Chart	53
6.4.7.	CCP System Software Indications	53
6.5.	SPARE PARTS.....	54

Figures and Tables

Figure 1-1 — CLM Nonredundant Five-Slot Module 1851	11
Figure 2-1 — CLM Rear View (w/CLMI or CCP only).....	15
Figure 2-2 — Typical CLM Port Connections.....	17
Figure 2-3 — Modem Usage from a CLM.....	18
Figure 3-1 — LCN I/O Board, Address Jumpers 2923.....	20
Figure 3-2 — K2LCN /K4LCN Address Pinning.....	21
Figure 3-3 — Cable for a Modem.....	23
Figure 3-4 — Cable for a Full Duplex RS485 Network	25
Figure 3-5 — Cable for a Half Duplex RS485 Network.....	25
Figure 3-6 — Cable for Redundant CLM	26
Figure 6.4.1-1 — CLMI Test and Status Indicators.....	48
Figure 6.4.4-2 — CCP Test and Status Indicators	51

1. INTRODUCTION

This section provides an overview of the CLM equipment and lists reference documents available from Honeywell.

1.1. GENERAL DESCRIPTION

The Communication Link Module (CLM) provides a method of economically connecting various RS-232 and 485 based devices to your Honeywell TPS system.

This manual guides you through the planning and installation considerations unique to the CLM equipment. However, this manual is not intended to be a substitute for the *LCN Site Planning* manual and *LCN System Installation* manual, listed in the standard LCN publications referenced in subsection 1.3 of this manual.

The CLM runs under a modified Application Module (AM) software personality. The hardware is similar to AM hardware, contained in a standard TPN Five-Slot or dual-node equipment module that supports the Local Control Network (LCN). In the CLM, the following boards and their appropriate input /output (I/O) boards are used to provide serial communication ports to RS-232 and RS-485 devices.

- Communication Link Module Interface (CLMI)
- Communication and Control Processor (CCP)
- Universal Communication Interface Module (UCIM)

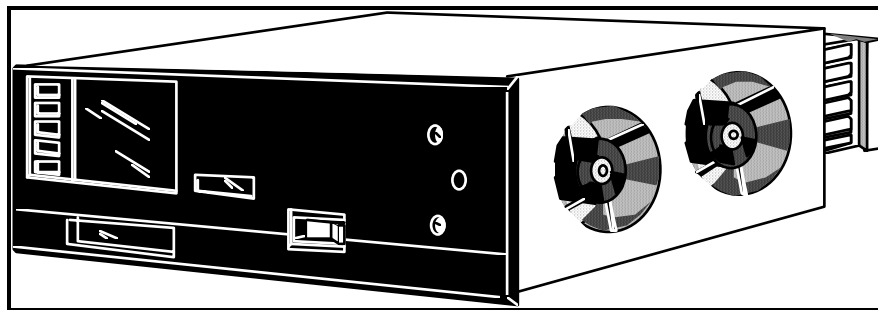


Figure 1-1 — CLM Nonredundant Five-Slot Module 1851

1.2. HONEYWELL SUPPORT SERVICES

Optional Installation Support, Field Services, and Technical Support are available during the on-site installation and checkout of TPS System equipment. Honeywell representatives are available to assist in interpreting this manual and to help resolve problems or situations not covered by this manual. A toll-free number (800-822-7673; in Arizona, 602-313-5558) is available for technical assistance.

1.3. RELATED PUBLICATIONS

Honeywell's standard documentation is included with the system. The following publications apply to the TPS System and should be referred to as required and available:

Title	Binder
LCN Site Planning	LCN Site Planning & Installation
LCN System Installation	LCN Site Planning & Installation
LCN System Checkout	LCN Site Planning & Installation
Application Module Specification and Technical Data	LCN System Summary
Five/Ten-Slot Module Service Manual	LCN Service - 1
Test System Executive	LCN Service - 2
Hardware Verification Test System	LCN Service - 2
Core Module Test System	LCN Service - 2
Maintenance Test Operations	LCN Service - 1
Process Operations Manual	Process Operations
CLM Planning, Installation, and Service (this publication)	CLM
CLM Platform Implementation	CLM
CLM Parameter Reference Dictionary	CLM
CLM Process Operations	CLM

2. SITE PREPARATION

This section provides the following information for the CLM: Storage conditions, site requirements, electrical requirements, and configuration information.

2.1. STORAGE CONDITIONS

If the CLM is to be placed in storage, follow these environmental constraints:

Temperature	-35 – +70°C
Humidity (RH)	10 – 80%, max wet bulb 20 – 30°C
Shock ¹	3.0 g for 10 milliseconds

Note that the temperature/relative humidity cannot be cycled such that moisture or condensation occurs on the equipment—keep the rate of change less than 6% per hour. These storage and shipping requirements are for one year duration only, provided the equipment is properly packaged and contains an adequate amount of desiccant (moisture removing agent).

2.2. SITE REQUIREMENTS

The CLM is designed for a Class A General Industrial environment. It must be operated in a temperature environment of 0 – 50°C (32 – 122°F). While operating, components of this system are not designed to withstand greater vibration than 1g.

This equipment is intended for use in a controlled environment. Although TPN equipment will operate at 0 – 50°C, Honeywell recommends a normal environment of 25°C (77°F) with a relative humidity of 40 – 50% to realize maximum life and enhanced reliability.

2.2.1. Dimensions and Weight

CLM — 5 Slot Module	Height	17 cm (7")	Weight	21.5 kg (46 lb)
	Width	48 cm (19")		
	Depth	61 cm (24")		
CLM — Dual Node Module	Height	17.78 cm (7")	Weight	18.2 kg (40 lb)
	Width	48 cm (19")		
	Depth	61 ccm (24")		
Dual Node Specs	Height	17.78 cm (7")		
	Weight	48 cm (19")		
	Depth	61 cm (24")		
	Weight	18.2 kg (40 lb)		

¹ When enclosed in the original shipping container.

2.2.1 Dimensions and Weight (continued)

Dual Node Specs	Height	17.78 cm (7")
	Weight	48 cm (19")

Dual Node Power		Avg	True RMS	Peak	Wattage	BTUs/hr
5 Slot Red	Typical	1.45	3.05	8.02	231	788
	Maximum	2.91	5.62	13.20	411	1027
Dual Node	Typical	0.40	1.04	3.12	67	228
	Maximum	0.61	1.44	3.96	87	297
5 Slot	Typical	1.05	2.33	6.38	168	573
	Maximum	1.60	3.31	8.20	219	747
Inrush current						
5 Slot	14A @ 120 Vac					
	28A @ 240 Vac					
Dual	6A @ 120 Vac					
	12A @ 240 Vac					

2.3. ELECTRICAL REQUIREMENTS

The ac power option, as selected by the customer, is installed in the CLM before shipment.

The inrush current to each electronics module is limited to 15A @ 120 V, 30A @ 240 V.

2.3.1. AC Voltage Options

120, 220, 240 Vac +10%, -15%

2.3.2. Frequency

47 Hz to 63 Hz

2.3.3. Current in Amperes at 120 Vac

		Avg	True RMS	Peak	Wattage	BTUs/hr
CLM — 68020 based, Non-redundant Module	Typical	1.05	2.33	6.38	168	573
	Maximum	1.60	3.31	8.20	219	747

CLM — 68020 based, Redundant Module	Typical	1.45	3.05	8.02	231	788
	Maximum	2.91	5.62	13.20	411	1027
CLM — 68020 based, Dual Node	Typical	0.40	1.04	3.12	67	228
	Maximum	0.61	1.44	3.96	87	297

Note: Specifications for the K4LCN and Enhanced Redundant AM were not available at the time of this printing.

2.4. CONFIGURATIONS (CLMI/CCP)

Both non-redundant and redundant CLM configurations are allowed. Depending upon certain limitations, the CLM can communicate either through modems or by a direct connection.

Figure 2-1 shows CLM connections. Refer to subsection 2.5.2 for examples of port cabling configurations.

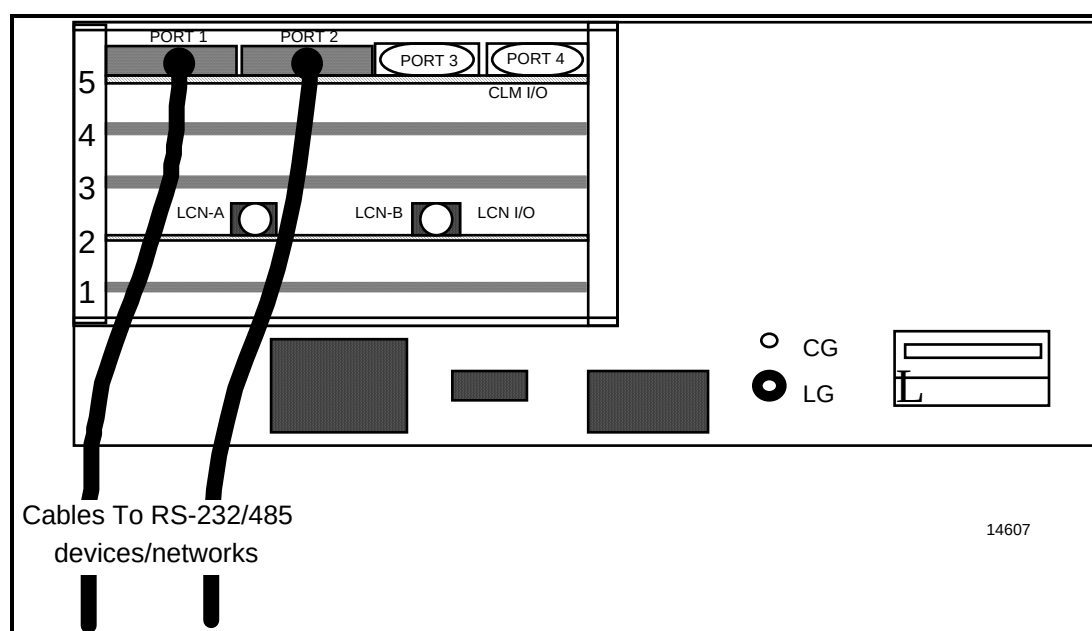


Figure 2-1 — CLM Rear View (w/CLMI or CCP only)

2.4.1. Cabling Considerations

Certain restrictions and options must be considered in planning your installation.

2.4.1.1 Cable lengths

The cables from the CLM ports cannot be longer than 15 cable-meters (50 cable-feet) when communicating RS-232. If the distance exceeds this limit, you must communicate 485 or use modems or electrical characteristics converters. Refer to subsection 2.5 for modem considerations. Subsection 3.2. shows how to wire the RS-232 cable supplied by Honeywell so that it interfaces a modem.

Important:

When connecting the field cables to the I/O paddleboard, it is important to tie the cables off in such a manner as to not place a strain on the I/O paddleboard. Connection ties or the LCN cabinet cable routing sleeve should be utilized.

2.4.1.2 Direct Connection

If you are connecting a single device to one of the ports, and the cable length is less than 15 cable-meters, you may use an RS-232 direct connection (no modems) cable.

In this arrangement, the RS-232 cable must be specifically wired to a connector which mates your device. Appendix A shows cable wiring schemes for other interface devices.

2.5. CONNECTIONS (CLMI/CCP)

2.5.1. Modem Usage and Selection

Either direct connection or modems can be used with the CLM. As mentioned earlier, direct connection is limited to a maximum of 15 cable-meters.

A short-haul modem presents an RS-232 hardware interface to the CLM similar to that presented by conventional telephone modems. The short-haul modem, however, uses separate lines (not telephone lines) and may take some liberties with interface protocol that would not be acceptable in conventional telephone modem communications.

Conventional telephone modems can be used with the CLM.

Broadband modems can not be used with the CLM.

Use short-haul modems over moderate distances (a maximum range of 2 to 4 kilometers). Honeywell has given limited exposure to these modem types and has found they performed adequately with the CLM—see Table 2-1 for an example.

Table 2-1 —Short-Haul Modem Examples

MANUFACTURER	MODEL NO.	DESCRIPTION
Black Box Co. Box 12800 Pittsburgh, PA 15241 (412) 746-5500	LD485A	64 drops maximum, 4-wire 26 AWG Twisted Pair Estimated maximum distance versus speed: 2.0 km (1.2 miles) @ 19.2 k baud 2.7 km (1.7 miles) @ 9600 baud 3.7 km (2.3 miles) @ 4800 baud

Some fiber-optic “modems” are available that may be used where the maximum distances above are insufficient. Please check with your Honeywell representative before utilizing this type of equipment in your application.

2.5.2. Non-redundant CLM Port Connections

Figure 2-2 (A) illustrates examples of directly connecting to a non-redundant CLM. A maximum of four devices can be connected in this manner.

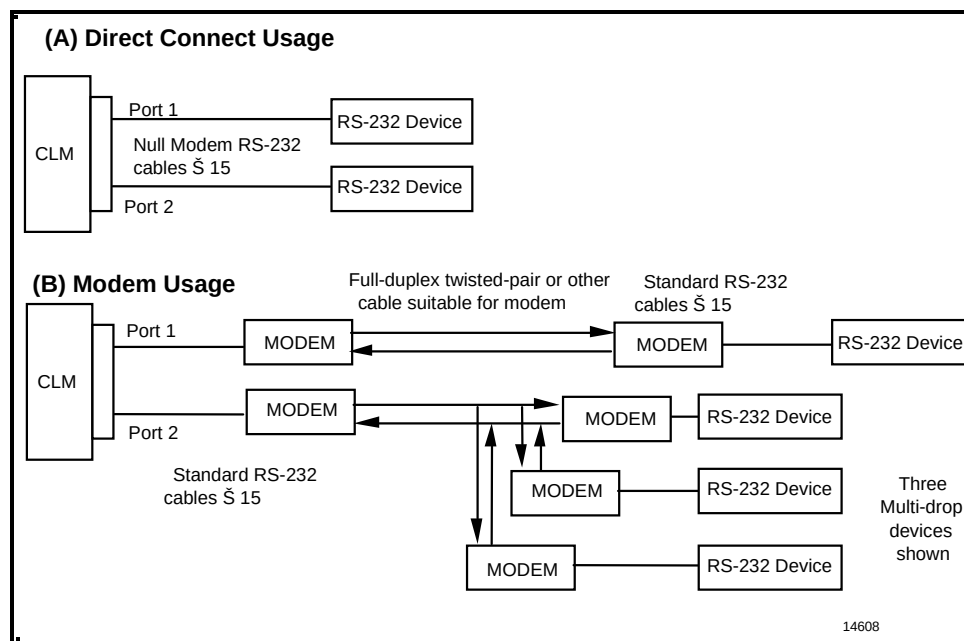


Figure 2-2 — Typical CLM Port Connections

Figure 2-2 (B) illustrates an example of modem usage from a CLM. The multi-drop connection to Port 2 in illustration (B) shows a typical “network” of devices. Additional loading caused by several modems on the network can reduce the overall length of the network twisted-pair cable; consult the manuals for your modem for particulars.

2.5.3. Redundant CLM Port Connections

Figure 2-3 illustrates the connections for redundant CLM ports.

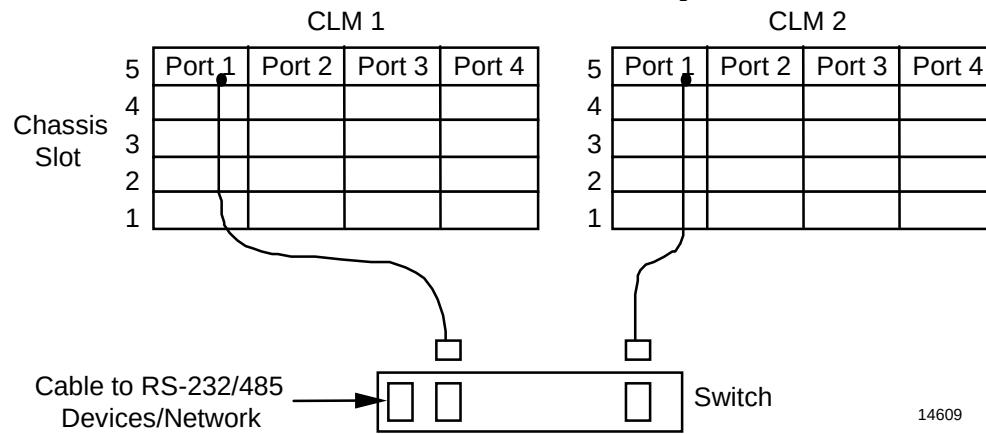


Figure 2-3 — Modem Usage from a CLM

3. HARDWARE INSTALLATION

This section provides information for unpacking and assembling the CLM.

3.1. UNPACKING

When the equipment arrives at the system site, open each shipping box, remove the protective wrapping, and carefully inspect each piece for any physical damage. If damaged, immediately notify the carrier and your Honeywell sales representative as to the extent and type of damage. Also check each piece of equipment against the invoice list for any missing items.

3.2. ASSEMBLY AND CABLING

Refer to Figure 2-1 for an illustration of a CLM installation.

NOTE

Some assembly and cabling of this equipment may have been done at the factory. If so, please check your equipment and verify that its installation is similar to that described.

3.2.1. Module Installation

Follow the procedure below to install the CLM module:

1. Remove the CLM module from its shipping carton.
2. Securely fasten the CLM module to its rack mount with the hardware provided.
3. Connect ground straps or leads from the base of the cabinet (or a nearby module) to the appropriate chassis-ground and logic-ground connections on the rear of the CLM module. Refer to the *LCN System Installation* manual in the *LCN Site Planning & Installation* binder for further grounding information.
4. Ensure the power switch on the front of the CLM module is off. Install the module power cord supplied. Refer to *LCN System Installation* manual in the *LCN Site Planning & Installation* binder for power wiring information.

3.2.2. LCN Node Pinning

Follow the procedure below to pin the node address on the CLM:

1. Power down the CLM module.
2. Remove the LCN I/O paddleboard from rear slot 2 in the CLM module. Refer to Figures 3-1 and 3-2 for instructions and the location of the pinning header on the LCN I/O board. Pin the node address as shown. Be sure to adjust the parity jumper as required.
3. Reinstall the LCN I/O paddleboard.

NOTE: Node addressing on the LCN I/O paddleboards is required on HMPU and HPK2 processor boards and optional on the K2LCN and K4LCN boards. If the node address is pinned on the I/O paddleboard when using a K2LCN or K4LCN processor, then all address jumpers must be removed from the processor board. Conversely, if the node is addressed on the K2LCN or K4LCN processor board, then all address jumpers must be removed from the LCN I/O paddleboard.

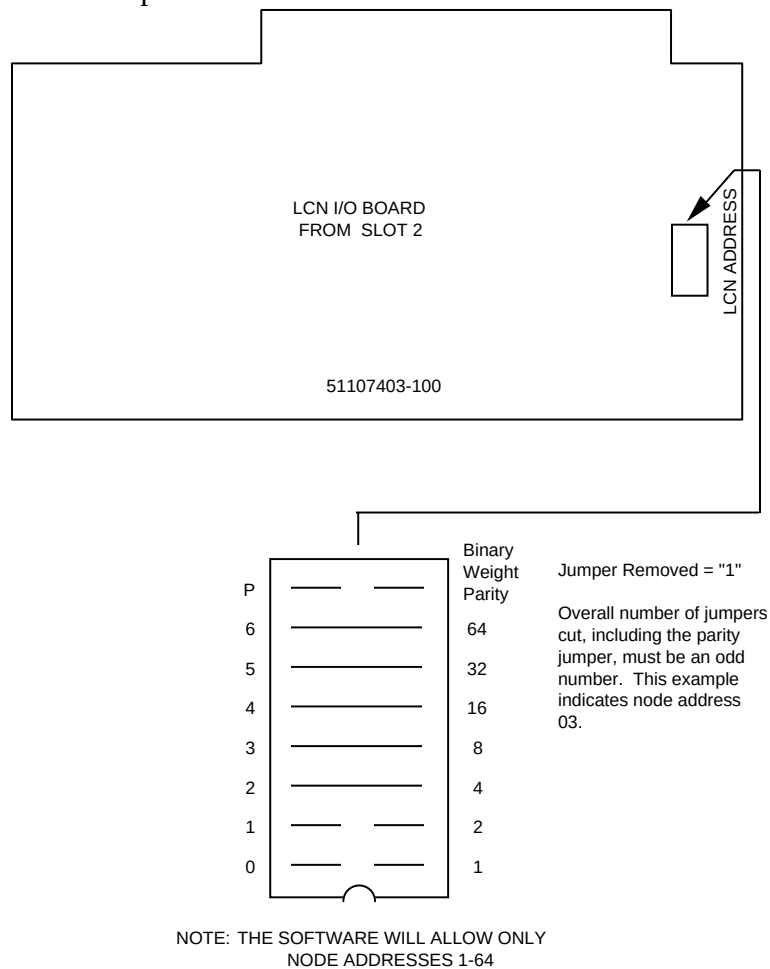


Figure 3-1 — LCN I/O Board, Address Jumpers 2923

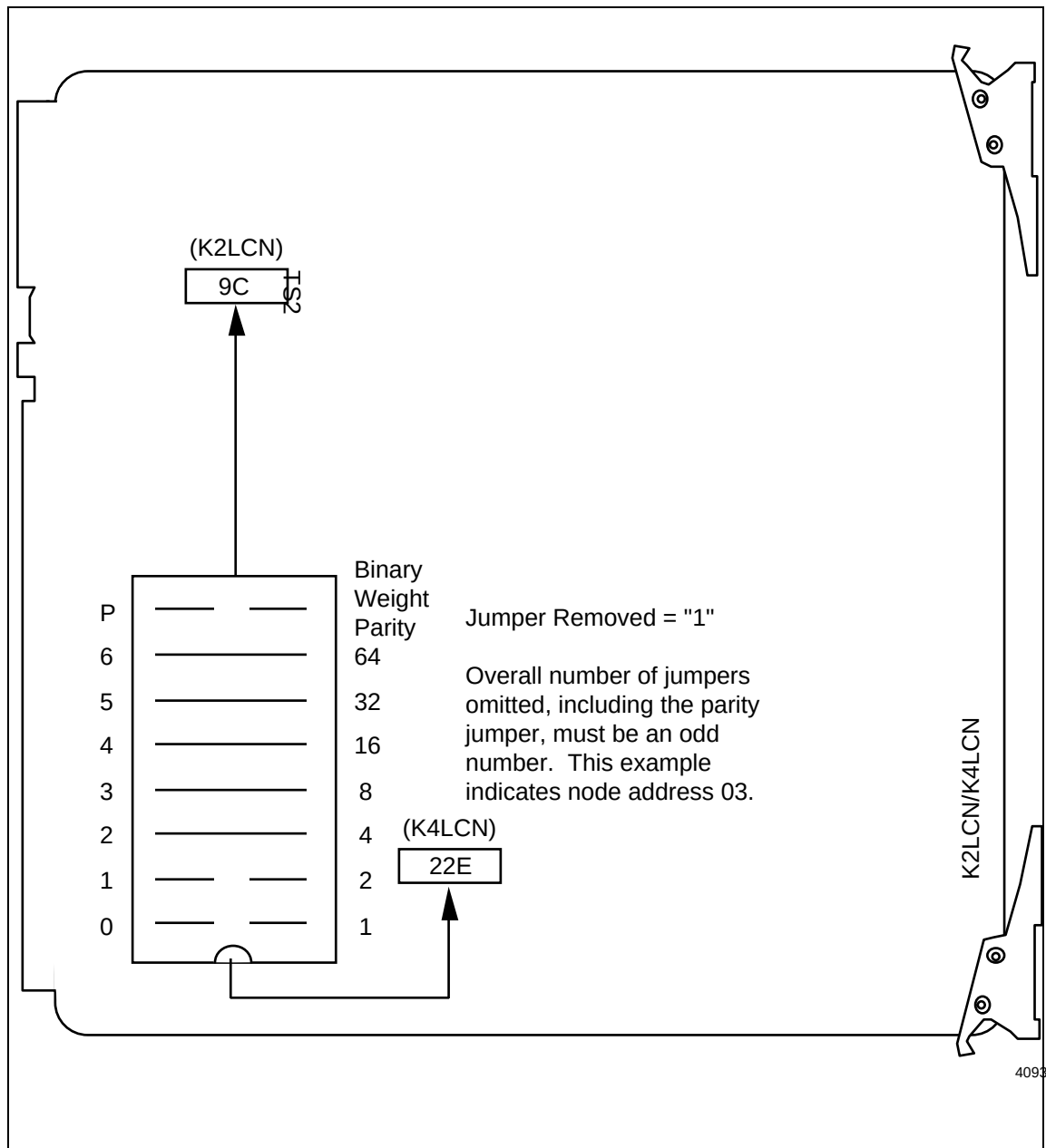


Figure 3-2 — K2LCN /K4LCN Address Pinning

3.2.3. CLMI, CCP, UCIM Pinning

These boards are pinned correctly from the factory and should not be changed.

3.2.4. CLMI/CCP 4-Port Interface Board

This assembly has three options for data transmission. Depending on the users configuration there are different pinning options associated with each port. Each port has two sets of jumpers (that is, Port 1 J5 RS-485 and J6 RS-232C) and can be configured individually.

3.2.4.1. RS-232C Pinning

RS-232C pinning requires all eight (8) jumpers in the RS-232C position. These jumpers are located above the 9 pin D-Sub connector (DB9) and are labeled J6 RS232, J8 RS232, J10 RS232, and J12 RS232. In addition to these eight jumpers there are four (4) suitcase jumpers used for terminating the RS-485 lines, (J13 through J16 four different positions). They **must not** be shorting the termination pins for the RS232C transmission.

3.2.4.2. RS-485 Pinning (prior to Interface Board Rev. E)

- RS-485 (2 wire) pinning requires all eight (8) jumpers in the RS-485 position. These jumpers are located above the 9 pin D-Sub connector (DB9) and are labeled J5 RS485, J7 RS485, J9 RS485, J11 RS485. The suitcase jumpers should be in the 2 wire termination position (2 W TERM J13 through J16, pin only the ports in use).
- RS-485 (4 wire) pinning requires J5 RS485, J7 RS485, J9 RS485, and J11 RS485 jumper #5 **open**. Jumpers 1 through 4 and 6 through 8 in the shorted RS485 position. The suitcase jumpers should be in the 4 wire termination position (4 W TERM J17 through J20, pin only the ports in use). This is the standard factory setting.

3.2.4.3. RS-485 Pinning (Interface Board Rev E. and greater)

- RS-485 (4 wire *AND* 2 wire) pinning requires J5 RS485, J7 RS485, J9 RS485, and J11 RS485 jumper #5 **open**. Jumpers 1 through 4 and 6 through 8 in the shorted RS485 position (**Note: Jumper 1 is closest to the DB-9 connector**). The suitcase jumpers should be in the termination position TERM J17 through J20, pin only the ports in use). This is the standard factory setting.

CAUTION

In redundant CLM applications using RS485, the TERM suitcase jumpers described above should only be in place for the interface board at the end of the cable. The board that is in between should have its jumpers removed.

P/N 51196019 - Supports RS232 communications only and **is** CE Mark Certified.

P/N 51305055 - Supports RS232/485 communications and **is not** CE Mark Certified.
Revision *E* or later is required for multi-drop 485 support.

3.2.5. Cabling and Pinning for Modems

NOTE

Cables should have been purchased and shipped with your CLM. If not, refer to the part numbers in the parts list table at the end of this document for cables necessary.

3.2.5.1. Modem Cable

For connection to a modem serving ports 1 or 2, as illustrated in Figure 2-2 (B) subsection 2.5.2, install cable 51196078 or 51196078-015.

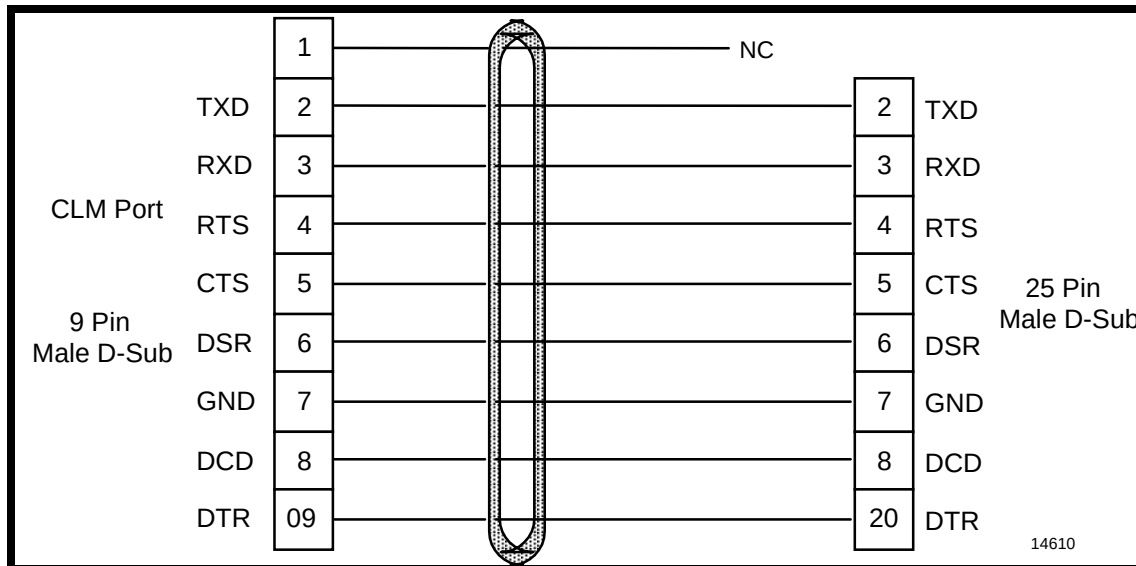


Figure 3-3 — Cable for a Modem

3.2.5.2. Modem Wiring

Use the instruction manual provided with your modems to install modem wiring between all of the modems. Be sure the wiring meets the following criteria.

- The multi-drop modem “telephone lines” must be 4-wire, full-duplex with the line driver of the local modem (CLM end) in parallel with the line receivers of all remote modems. Likewise, the line drivers of all remote modems are in parallel with the line receiver of the local modem.
- A single pair of modems (point-to-point) must also be 4-wire, full-duplex with the line driver of each modem in parallel with the line receiver of the other modem.

3.2.5.3. Modem Pinning

Use the instruction manual provided with your modems to properly pin your modems. Be sure they meet the following criteria.

- The local (CLM end) modem must be pinned to hold its transmitter enabled at all times, normally by putting RTS/CTS in the ON position.

The local modem must also be pinned to hold RXD in the marking state in the absence of a valid carrier.

- Each remote (CLM end) multi-drop modem must be pinned to enable its transmitter only when the device raises RTS.

All remote modems must also be pinned for an RTS/CTS delay of approximately 5 to 15 milliseconds. This delay can be determined empirically by the user, based upon the hardware chosen. We have found the LD485A Black Box modems (Table 2-1) operate satisfactorily using the 5 millisecond delay.

- RTS/CTS delay need not be pinned in the local modem of a multi-drop network, and is also immaterial in a point-to-point arrangement.

3.2.6. Cabling for RS485 Networks

3.2.6.1. RS485 Port Connections

RS485 networks may be connected to the CLM Interface board port using a 2-wire or 4-wire configuration. The Port pin connections are listed below.

RS485		
4-Wire (Full Duplex)		2-Wire (Half Duplex)
1 - n/c		1 - n/c
2 - TX+		2 - TXRX+
3 - RX+		3 - TXRX+
4 - TX-		4 - TXRX-
5 - RX-		5 - TXRX-

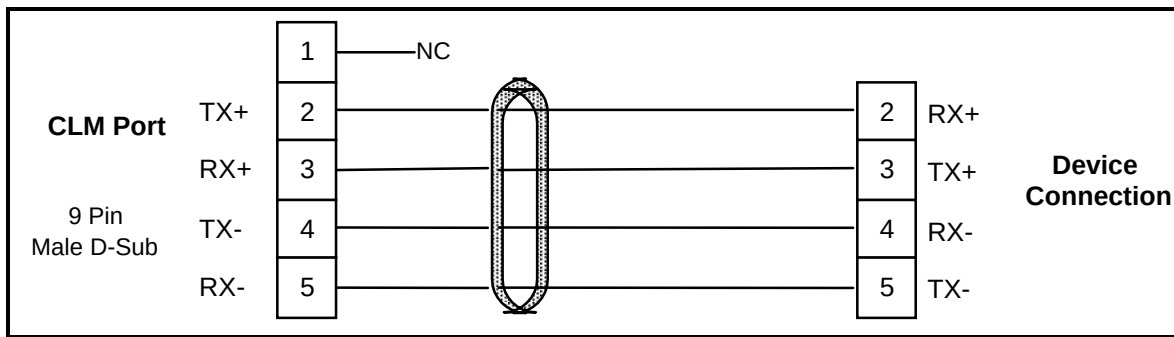


Figure 3-4 — Cable for a Full Duplex RS485 Network

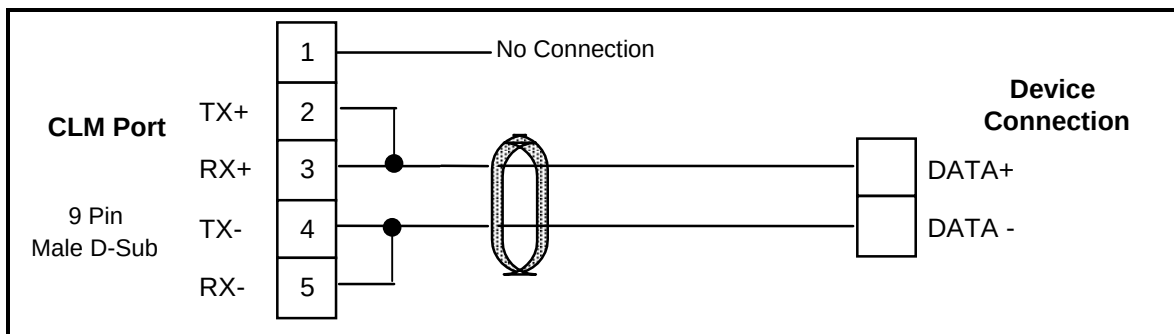


Figure 3-5 — Cable for a Half Duplex RS485 Network

Connections from Redundant CLMs to RS485 networks is supported by 4-wire (Full Duplex) only. Redundancy is supported by CLM Interface Boards with revision *E* or greater and CLM I/O boards (assembly # 51305055-100) with revision *D* or greater.

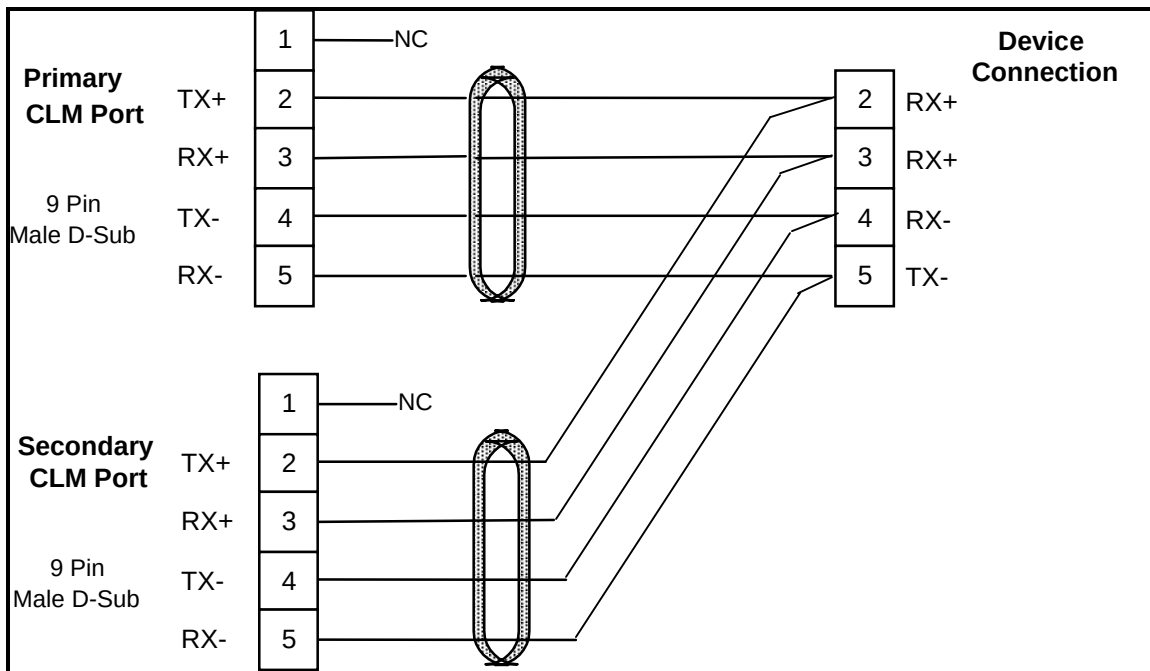


Figure 3-6 — Cable for Redundant CLM

3.3. INSTALLATION WRAP-UP

Dress all cables neatly and out of the way to protect them from accidental damage. Plug all modems and CLM modules into their proper receptacles.

4. CHECKOUT

This section tells you how to check the CLM after it has been installed, plugged in, and is ready to go.

4.1. POWER-ON TESTS

You do not need to have the RS-232 device installed or connected to perform the checks in this section. Do not, however, perform any of these tests until all other parts of the CLM have been installed according to Section 3.

4.1.1. Electronics Checks

Perform these electronics checks on the CLM electronics module.

1. Remove the front cover of the CLM module. Before turning the power on, ensure the **LO-NOM-HI** jumper or switch on the front of the power supply is in the center or **NOM** position.
2. Set the **POWER** switch to **ON** while observing the Light Emitting Diodes (LEDs) on the power supply, fan assembly, and on the individual cards in the unit.

Note that the red LEDs on the boards light for a few seconds (it takes less than 30 seconds to complete the power-up tests), then they turn off and the green LEDs turn on. If any red LEDs on the boards remain on, some portion of the power-up tests have failed—record the alphanumeric status display code and proceed to the *Five/Ten-Slot Module Service* manual in the *LCN Service - 1* binder.

If the power-up test is successfully completed for all nodes, all green LEDs on all boards are on (there may also be some yellow LEDs on or flashing), and the alphanumeric status display indicates the node address you set in subsection 3.2.2 of this manual.

3. Check the power supply status LEDs (**POWER OK** and **ERROR**) and the **FAN ALARM** LED. Note that the **FAN ALARM** and **ERROR** LEDs are off and the **POWER OK** LED is on (it is a fault condition if both the **ERROR** and **POWER OK** LEDs are **ON**).
4. Press the momentary **RESET** switch. Note that the power-up tests are initiated similarly to Step 2, and the results are satisfactory.
5. Replace the front cover.
6. Proceed with the CLM Platform Software Installation, section 5.

5. CLM Platform Software Installation

5.1. Introduction

5.1.1. AM-based Installation

There are two versions available for CLM/Platform Software installation.

- **CLM Platform** is the original CLM, which is installed to an AM node and requires the CLM communications boards, CLMI, CCP, or UCIM.
- **TPS/CLM Platform** installs to the APP node with an AM personality installed and emulates the previous CLMI/CCP communications boards used in the earlier version of the CLM.

This manual describes **CLM Platform** software installation to an AM with CLM communications boards. Refer to the *TPS/CLM Planning & Installation Manual* for TPS installation.

5.1.2. Configuration overview

The CLM is configured in an AM node by:

- Installing CLM Interface Boards;
- Adding CLM Platform software;
- Building required LCN support points;
- Building CLM operating displays;
- Adding required information for the node in the NCF (Modify); and
- Adding one or more applications by:
 - establishing the physical port connections,
 - configuring the non-TPN network,
 - building the application in the non-Honeywell devices,
 - configuring AM-based points to image the non-Honeywell controller points, and
 - building the point interface table structures to connect the non-Honeywell controller points to these image points.

The following subsections describe the procedures required to accomplish the platform installation outlined above.

Before performing the platform installation, you need to understand the implication of *unit names*. This is described in Section 14 of the *Engineer's Reference Manual*, located in the *Implementation/Startup and Reconfiguration - 2* binder.

5.2. Install CLM Platform Hardware - Summary

5.2.1. Procedure

For each CLM Node in your system, install and/or connect the required hardware.

5.2.2. Hardware Installation

Step	Action
1 (Platform Specific)	Install the Honeywell-supplied CLM Interface Boards in the Application Module that is designated as the CLM Node, if required. Note: You can install 1 or 2 boards per CLM, depending on your system requirements.
2 (Application Specific)	Connect the configured ports on each board installed to your non-Honeywell network/devices. Note: You can connect from 1 to 4 ports per CLM Interface board, depending on the board type (that is, CLMI, CCP, UCIM) and your system and application requirements.

5.3. Install CLM Platform Software

5.3.1. Before You Do the Software Installation

- 1) Create a directory in one of the destination user volumes.
 - The directory name of “UCLM” is suggested and used in this procedure.
 - This directory must have a minimum of 15 megabytes of disk space available and a maximum file setting of at least 2000.
- 2) From the **Modify Volume Paths** display, set the path for:
 - CL SOURCE/OBJ to NET>UCLM>
 - CL PARAM LIST* to NET>UCLM>
 - USER DEFLT PATH* to NET>UCLM>
 - *(same path as the CL SOURCE/OBJ).
 - CL CUSTOM GDF to NET>CDSG>

(verify the directory CDSG exists)
- 3) From the Command Processor, verify that:
 - &CUS and &CLX directories also exist (this was completed during the installation of the Application Module software)
 - Maximum Files is at least 2000 and free disk space is at least 15 m bytes.

Note:

Some files included in the product release are copied to your system wherever specified within the install process.

Source files for custom displays are included and should be put into directories included in the schematic search rule (UCLM is suggested.).

The following table is a summary of the CLM Platform software installation.

Step	Description
1	Execute the provided install EC file and check for errors
2	Build a CLM Network Status Point and a Builder Point for each CLM Node installed on the system
3	Customize and compile CLM Platform operating displays
4	Define the configuration required for communications

5.3.2. Execute the Install Procedure

In preparation for executing the install procedure:

- Read the Customer Release Guide, supplied with the package, for information that applies to your installation including HM sizing requirements, special instructions, and reinstallation information.
- Mount the CLM Platform release media.
- Visually scan the file - INSTALL.EC - provided on the supplied media, for an overview of the install process.

Important – review the CDS packages to be installed and compiled to assure **no conflict** with existing CDSs on the system, that is, no duplicate names.

The install EC file performs the following functions:

- checks for previous installation of the package.
- prompts for the current TPS release for loading of the specific release dependent files to the HM. Some files to be loaded are independent of the LCN release other files must be loaded or compiled only for a specific release.
- prompts for confirmation of load path (directories) setup.
- copies required files from the supplied media to the designated HM directories.
- compiles supplied CDSs required for package operation.
- creates a file, INSTALL.LS, of the install execution for verification of correct installation, no errors listed, and for future reference.

Do the following 3 steps to complete the EC installation:

STEP 1: Initiate the EC File	In the Command function, within the Engineering Personality, enter the following command to begin the install procedure: EC \$Fn>&CLM>INSTALL.EC \$Fn where n = the logical device identification where the CD is mounted (e.g. \$F1)
STEP 2: Follow the Prompts	Follow the prompts output by the install execution and enter the required information to continue through the procedure. Caution: An incorrectly formed pathname entry may cause the INSTALL script to abort prior to completion. If this occurs, the INSTALL script may be restarted by setting the prompt data off (type PD OFF and press ENTER) and closing the verification output file (type DO and press ENTER), then re-executing the INSTALL.EC.
STEP 3: Verify the Results	Print out the INSTALL.LS file and confirm that there were no errors during the execution of the EC file.

5.3.3. Build Required CLM Platform Points

5.3.3.1. Build the Required CLM_nn & BUILD_nn LCN Points

A CLM Network Status Point and a Builder Point are required for each CLM node configured. These points provide alarming and display support to the CLM Platform. These points can be generated by modifying the EB files included in the release as explained below.

From the Command Processor, **do** the following steps

Step	Action
1	Set the current pathname to NET>UCLM>.
2	- Copy the CLM_NN.EB file to another .EB file following CLM naming convention where NN is the appropriate CLM Node number (for example, CLM_55.EB). - Edit the new file, changing the point name, unit, and optionally the description, and keyword to the desired values. Notes: The processing cycle (period) has a default value of 30 seconds, but can also be changed. The CDS packages attached to the point are NETCDS, NETCDS1, NETCDS2, NETCDS3, NETCDS4 These were compiled in the execution of the INSTALL.EC task. They are required for the CLM_NN point.
3	- Select the BUILDER COMMANDS from the Engineering personality menu. - Select EXCEPTION BUILD - Select the Load Entities option with your .EB file identified as the input using CLM_NN as the IDF file name. - Initiate an exception build of the point into the CLM/AM database.
4	- Copy the BUILD_NN.EB file to another .EB file following the CLM naming convention where the NN is the CLM node number (e.g., BUILD_55). - Edit the new file, changing the point name, unit, and optionally the description and keyword to the desired values. The CDS packages attached to the point are BUILDCDS, BLD2CDS, CLMDCDS These CDSs were compiled in the execution of the INSTALL.EC task. They are required for the BUILD_NN point.

Step	Action
5	• Select the Builder Commands from the Engineering personality menu. Select EXCEPTION BUILD • Select the Load Entities option with your .EB file identified as the input using BUILD_NN as the IDF file name. • Initiate an exception build of the point into the TPN database.
6	• Call the Detail display of the new BUILD_NN point. Set the initial default pathnames by entering appropriate text into the following string parameters. • Move to the Custom pages and change the following parameters. ENTARR(101) = &CUS <Default = CLM Save Data > ENTARR(102) = NET>&CUS <Default = CLM Builder Support > ENTARR(103) = NET>&CUS <Default = CLM Application > Note: After the CLM custom displays are compiled in the next step, these pathnames may be changed or viewed from the CLM CONFIG DATA page.
7	• Now checkpoint the CLM: - Go to the Application Modules Status Display, - Select the CLM/AM node, and - Select SAVE DATA to checkpoint to the HM (default destination).
8	Repeat Steps 2 through 7 for each additional CLM.

5.3.4. Customize Displays

5.3.4.1. Custom Graphics Must be Compiled

The custom displays required by the CLM Platform are included in the release media and were copied to the HM in the previously executed INSTALL.EC process.

The CLM Status Display "CLMSTAT1" must be modified to include the CLM node(s). If other CLMs are added to the LCN later, they must be added to the CLM Status Display by following this same procedure.

Do the following steps to customize the displays.

Step	Action
1	From the Engineering Menu start the Command Processor. set the current path to the directory where the CLM displays are stored (usually <i>NET>UCLM</i>) by entering: SP NET>UCLM>
2	Invoke the Picture Editor by entering the command: PE
3	Load the Global Database names by entering the command: LD CLMDDB
4	Read in the display by entering the command: R CLMSTAT1
5	Modify the display by entering the command: SEL VAR;MOD;
6	At the top of the display are 20 sub-pictures in two columns indicated by the text: <i>SUB CLMID1</i> Select the first sub-picture text in the left column. Then press <i>ENTER</i> .
7	Make the following change in the variant text. Change: IF 1 <> 1 THEN "SUB CLMID1" To read: IF 1 = 1 THEN SUB CLMID1 NOTE: Remember to remove the “ “ marks. Then press <i>ENTER</i> .

Step	Action
8	A screen appears that allows you to enter information about a CLM: ENTER THE LCN CLM POINT ID Enter the name of the CLM point built previously (CLM_nn). ENTER Node Number (e.g. 49) Enter the LCN node number (01-96). ENTER Redundant Node Number If the CLM is redundant, enter the backup node number (nn+1)
9	Repeat steps 5 through 8 for each CLM to be added to the display.
10	Compile the display by entering the command: COM Press ENTER to overwrite (save).
11	All of the other custom displays included with the CLM must now be compiled. The Picture Editor Multiple Compile (MCOMP) command is used to compile all required displays. Enter: LD CLMDDB MCOMP NET>UCLM>CLM_DS Check the CLM_DS.ER file for compilation results.
12	Exit the Picture Editor by entering: END

ATTENTION
Important Notes!

The schematic "CLMSCHEM" is the initial display in the CLM Network Display process. Its function is to establish the required values in the Honeywell variable area.

When this display is read in and compiled, *it will look like a blank screen*. In fact, it does have a process included in the "Initial" target attached to the display.

The schematic "CLMSCHEM" is the lead display and can be put on a button, if desired. It can also be accessed using the "SCHEM" key or from the "VIEW OBJECT DETAIL" target of one of the CLM objects on the status detail display. Refer to the *CLM Process Operations Manual*.

Update the area database with the pathname to the graphics, if required.

Be sure that the unit with the CLM_nn and BUILD_nn points is assigned on the console where the CLM displays are used.

5.3.5. Configure the CLM Interface Board (CLMI,CCP,UCIM)

5.3.5.1. You Must Define the Board Physical Location and Interface

This step defines the configuration information required by the platform to identify what hardware and software is used by the application to accomplish communications with the external devices.

The configuration information resides in a specific file for each different CLM node (or redundant node pair) on the LCN (sample file CLMI nnn .ZC). This CLMI configuration file is an ASCII text file with the following characteristics:

5.3.5.2. The CLMI nnn .ZC File

- CLMI nnn .ZC
where nnn = CLM physical node number (000-096) or the lower of the node numbers of the redundant pair
- Located in the directory of NET>&CUS> by default or the configured path indicated on the CLM Configuration page

5.3.5.3. ZC File Directives

- The following are the supported configuration directives:
CLMI <board #>, <slot #>
LOAD <file name>
INIT <0>, <port map>, <arg#3>, <arg#4>, <arg#5>
RUN <entry offset> (CLMI and CCP only)
END
- One set of operating system directives (CLMI, LOAD, RUN, and END) are required for each CLM Interface board configured.
- One set of protocol specific directives (LOAD, INIT) is required for each protocol application to be downloaded to the board.

Do the following steps to create a valid configuration file.

5.3.5.4. Edit the Configuration File

- Copy CLMI nnn .ZC to a new file, replacing nnn with the actual three digit node number.
Example file name: NET>&CUS>CLMI024.ZC
- Invoke the LCN Editor and read in the new ZC file.
- Determine and enter the logical board number and physical chassis slot location.
Note: You can use the Maintenance Personality on the User Station to determine the Interface board slot numbers. This is available through the SMCC/MAINTENANCE, REV/CONFIG status display for the target node. Note that all Interface boards will be defined on this display as PLCI.
- Define the correct operating system for the board.
(See the example below for directive values)

5.3.5

- Enter the application executable files (*.S) into the directives and define the ports to be used.
(See the example below for directive values)
- Repeat for a second board, if applicable.
- Save the file and exit.

5.3.5.5. ZC File Example

The following is an example of an application configuration file that uses two interface boards; a CLMI in physical node slot 5 and a CCP in physical node slot 4. The specific applications used are defined as APP1 and APP2. Note that the asterisk (“*”) is used to place comments in the file.

```
*
* EXAMPLE CONFIGURATION FILE (.ZC)
*
* FILE NAME: NET>&CUS>CLMI024.ZC
*
CLMI 1,5                (Board #1 in slot #5)
LOAD CLMI.S             (CLMI operating system)
RUN 41100               (CLMI operating system)
LOAD APP1.S             (Protocol Specific)
INIT 0,11000000         (Protocol Specific)
END
CLMI 2,4                (Board #2 in slot #4)
LOAD CCP.S              (CCP/UCIM operating system)
RUN 7F001100            (CCP/UCIM operating system)
LOAD APP2.S             (Protocol Specific)
INIT 0,00110000         (Protocol Specific)
END
```

Note: The LOAD APP1/APP2 and INIT directives are examples of the protocol specific directives that are added/included when the specific protocol is installed.

ATTENTION

The LOAD CLMI.S and RUN 41100 directives would be replaced with a LOAD CCP.S and RUN 7F001100 respectively if the CCP or the UCIM board is used rather than the standard CLMI board.

The following table shows the Operating System file and RUN address for each interface board type:

Board	Part Number	Description	Operating System Files	RUN Address
CLMI-P	51006877-100	CLMI (256 Kbytes)	CLMI.S	41100
CCP-1	51402426-200	CCP (1 Mbytes)	CCP.S	7F001100

5.4

Board	Part Number	Description	Operating System Files	RUN Address
CCP-4	51402426-100	CCP (4 Mbytes)	CCP.S	7F001100
UCIM-MP	--	UCIM - Modbus Plus	CCP.S	7F001100
UCIM-LP	--	UCIM - SG100	CCP.S	7F001100

5.4. Update LCN Network Configuration File

5.4.1. NCF update process

The LCN Network Configuration File (NCF) must be updated to identify each AM that is configured as a CLM.

The two steps for safely updating the LCN Network Configuration File (NCF) are:

Step	Description
1	Setting up for backup of the existing NCF to removable media.
2	Updating the NCF for each new CLM and error checking.

5.4.2. CLM is External Load Module (.LO)

The standard TPN Custom Loader uses the Network Configuration File to load the CLM Platform software.

Note: Complete information regarding NCF updates is included in the TPN Implementation/Startup & Reconfiguration - 1, Network Data Entry manual (SW11-305). The following is a summary.

5.4.2.1. Always Backup NCF Before Making Changes

Set up to back up the existing NCF to removable media. Always copy an existing NCF to removable media before installing a new NCF. This allows system recovery without restarting nodes should the new NCF cause problems in loading nodes or the HM autoboot after a power failure.

5.4.2.2. Setup to Back Up Existing NCF to Removable Media

Step	Action	Result
1	Select the SUPPORT UTILITIES target from the Engineering Personality Main Menu, with Keylock in the ENGR position.	The Utilities menu displays.
2	Select the MODIFY VOLUME PATHS target.	The Modify Default Volume Path Names Menu displays.
3	Insert the removable medium that will contain your backup NCF into a drive with a known designation.	
4	Find the display entry port for NCF BACKUP PATH.	
5	Enter the drive and volume name of your backup NCF removable media. The format is \$Fn>&ASY> where n = drive number containing the backup NCF – for example, \$F1>&ASY>	The backup pathname is established.
6	Verify that the port for EXT LOAD MODULE has NET>&CUS> in it. Then press ENTER.	The existing NCF is now ready to be backed up to removable media. Actual backup is automatically initiated DURING the NCF update process.
7	To return to the Engineering Personality Main Menu, select MAIN MENU .	Engineering Personality Main Menu displays.

5.4.3. Modify NCF

Update the NCF for each new CLM and check for errors. During the processing, the system automatically makes a working copy and a backup copy of the existing NCF; then it updates the working copy. If this process is successful, the system then updates the backup copy also.

This procedure must be repeated for each CLM Node in your system.

Step	Action	Result
1	Select the LCN NODES target, from the Engineering Personality Main Menu, with Keylock in the ENGR position.	This invokes the Network Configurator and displays a list of all the LCN Node numbers presently configured.
Note: Make sure the words ON LINE appear as the indicated mode (second line from the top). If they do not, refer to the <i>Network Data Entry</i> manual in the <i>Startup and Reconfiguration - 1</i> binder to find out why you are not in on-line mode.		
2	Select the node number of the AM to be configured as a CLM.	
3	Select the MODIFY NODE target on display page 1.	Page 2 is displayed.
4	Change the number of Background CL tasks as needed (not required for normal CLM operation). Caution: The CL Background tasks are not required for normal CLM operation. Care must be taken to not exceed the total number of tasks available. The CLM uses 9 of these tasks, Data Access uses 4, and certain AM external load modules may use 1. Prior to LCN R430, this cumulative value must be ≤ 20 . For LCN R43x, this cumulative value must be ≤ 25 . For LCN R500 and later, this cumulative value must be ≤ 30 .	
5	Change the Concurrent Data Access" value to 4. See caution in step 4.	
6	Change the REDUNDANT BUFFER INCREASE (# of 32 KW blocks) to at least 10, if redundant CLMs are implemented. Leave at 0 if non-redundant.	
Note: Some included CL modules execute in the background insertion order. If these CL blocks are used, then the CL Background tasks must be set to an appropriate value and factored in to the total number of tasks calculated.		

5.4.3

Step	Action	Result
7	Page forward to page 3.	Page 3 displays.
8	If the module CONV is not already included then it must be put in the 1st slot/entry location using the personality of AMO. In the next available slot/entry, add the module CLM with the personality of AMO. After these modifications are entered, press ENTER to effect the changes.	The indicated mode at the second line from the top is replaced with the indication that a modification is pending ("MOD AM ###" in white).
Note: Configuration changes must be checked (by pressing the F1 key) and installed (by pressing the F2 key and then pressing ENTER). The Network Configurator does not allow any other changes until you either install the data, OR abort the change by pressing the F5 key. Therefore, complete this procedure by performing the following steps.		
9	Press the Check Key (F1) to check the syntax of the configuration update.	If the syntax check is successful, the AM/CLM Node name (in the mode indicator location on the second line) changes from white to yellow. If the syntax check fails, it changes to red.
Note: If the F1 check fails, correct and repeat the step.		
10	Press the Install Key (F2) and then press the ENTER key to install the configuration update.	The NCF update is installed.
Note: Wait for the INSTALLATION COMPLETE message. The existing NCF file is copied to removable media. The working copy becomes the new NCF.		
11	Press CTL/MENU to complete this step and terminate the NCF updating.	The update is complete.

ATTENTION

Reference Section 9 in the *CLM Platform Implementation Manual* for CLM Redundancy/Automatic Failover NCF information.

5.5. Load the AM with CLM Platform Software

5.5.1. Restart the node

This procedure must be performed for each AM/CLM Node installed or modified.

5.5.1.1. Load CLM Platform Software

Step	Action	Result
1	Shut down the AM. If redundant, then both nodes MUST be shut down.	
2	Restart the AM/CLM Node.	The CLM Platform software is loaded as part of the standard AM load. Upon completion of the software load to the CLM Node, the Custom Loader activates the CLM Startup task. (Refer to Appendix C.) When the CLM Startup task concludes its functions, the CLM Node is in a normal operating state.
3	Check the journal entry file for errors.	If the CLMI configuration file is incomplete or was not accomplished previously, an error condition is identified in the operator messages entry file.

At this point, the CLM platform is installed and any further installation process will be application specific.

To install an application, the platform functionality BUILDER is required. Now that installation is complete, BUILDER is available for use.

This is the end of CLM platform-specific installation.

Proceed to the protocol specific user's manual to install the Communication Protocol and the Network Interface Layer of the CLM.

5.6. Add Other CLMs After the First Installation

5.6.1. Adding other CLMs

Only a portion of the installation procedure is required. Start with Build Points in section 5.3 and perform all subsequent procedures up to and including the procedure to Load the AM with CLM Platform Software, section 5.5.

5.6.2. Summary procedure

Add other CLMs after the First Installation.

Step	Action
1	Build the CLM_NN and BUILD_NN points.
2	Checkpoint the CLM.
3	Modify and compile the CLM Status display/schematic.
4	Build the required CLMI directive file.
5	Modify the NCF parameters.
6	Restart the CLM/AM Node.

6. SERVICE

This section presents service instructions unique to the CLM.

6.1. OVERVIEW

The CLM is housed in a standard TPN five-slot or dual node module. A unique functional circuit board, an input/output card and special cabling have been added to this standard module.

This manual provides instructions to test, troubleshoot, and repair those components unique to the CLM. Troubleshooting, disassembly, and assembly procedures for the remaining five-slot module and its components are contained in the *Five/Ten-Slot Module Service* manual in the *LCN Service - 1* binder.

Examples of different board configurations are shown below. Your CLM node may or may not utilize one of these example configurations. The important rules to remember are:

- The AM/CL node must contain a minimum of 6 megawords memory.
- There may be 1 or 2 CLMI/CCP, or UCIM boards per node.
- All LCN boards must be at the minimum LCN AM-based revision levels.

The following are examples of board complements for the 68020-based CLM:

Slot	Front	P/N	Rear	P/N
5	CLMI-P	51196877-100	CLMI I/O	51305055-100
4	EMEM	51400910-100		
3	EMEM	51400910-100		
2	LLCN	51401291-100	LCN I/O	51107403-100
1	HPK2-2	51401288-100		

An example of an alternate setup using the CCP board is specified below.

Slot	Front	P/N	Rear	P/N
5	CCP	51402426-100	CLMI I/O	51305055-100
4	blank			
3	EMEM	51400910-100	LCN I/O	51107403-100
2	LLCN	51401291-100		
1	HPK2-3	51401288-200		

An example of an alternate K2LCN setup is specified below.

Slot	Front	P/N	Rear	P/N
3	blank			
2	CLMI-P	51196877-100	CLMI I/O	51305055-100
1	K2LCN-4	50401551-400		

An example of two CLMI boards configured is specified below:

Slot	Front	P/N	Rear	P/N
5	CLMI-P	51196877-100	CLMI I/O	51305055-100
4	CLMI-P	51196877-100	CLMI I/O	51305055-100
3	EMEM	51400910-100		
2	LLCN	51401291-100	LCN I/O	51107403-100
1	HPK2-	51401288-200		

The following is an example of configuration if redundant AMs are used for CLM automatic failover:

Slot	Front	P/N	Rear	P/N
5	CLMI-P	51196877-100	CLMI I/O	51305055-100
4	QMEM-2	50401072-200		
3	AMR	51401070-100	Red I/O	51304159-100
2	LLCN	51401291-100	LCN I/O	51107403-100
1	HMPU	51400978-100		

The following are restrictions for 68020 redundant CLM configurations:

- Only 1 CLMI, CCP, or UCIM can be used in a 68020 redundant CLM.
- The HMPU processor board is required in the 68020 redundant AM/CLM.
- This configuration **MUST** be used for LCN software releases 3xx and 4xx and **MAY** be used in R5xx.

The following are restrictions for K4LCN redundant CLM configurations:

- The Enhanced AM Redundant Module (EAMR) must be used.
- Only 1 CLMI, CCP, or UCIM can be used in a normal K4LCN redundant CLM that uses two dual-node chassis modules.
- Two CLMI, CCP, or UCIM boards may be used if two five-card chassis modules are used.
- This configuration cannot be used in LCN software releases R3xx or R4xx.

6.2. FIELD ADJUSTMENT

There are no hardware field adjustments for the CLM. When replacing a board, do not alter pinning.

6.3. GENERAL TROUBLESHOOTING

Before investigating deeply into a problem, make some preliminary checks:

- Is power applied to the module? Check switches, fuses, and circuit breakers on all equipment to insure they are functioning.

WARNING

DO NOT REMOVE OR REPLACE CIRCUIT BOARDS WITH THE POWER ON.

Do not remove, handle, or transport circuit boards without observing proper Electrostatic Discharge (ESD) procedures. To review ESD procedures, see the *LCN Site Planning* manual in the *LCN Site Planning & Installation* binder.

- Note that the functional boards can be accessed through the front of each module by removing the front cover. Inspect the confidence indicators on each processor (both the AM processor board and the CLMI or CCP board). Check power supply and fan confidence indicators. Double-check the pinning on the CLMI board (shown in subsection 3.2.3).
- The small I/O or paddleboards are accessed from the rear of the module.

To isolate a failed board, power supply, fan assembly, or other Optimum Replaceable Unit (ORU), follow the service procedures provided in the *Five/Ten-Slot Module Service* manual in the *LCN Service - 1* binder.

6.4. CLMI and CCP TROUBLESHOOTING

The CLMI and CCP boards have unique indicators on their front edge to offer confidence that the board is working, and to provide assistance in case of a failure. The board also communicates with the Universal Station X (U^XS) to report software indications of hardware failures. This section explains the function of these hardware/software indicators and guides you in finding a failure.

6.4.1. CLMI Hardware Indicators

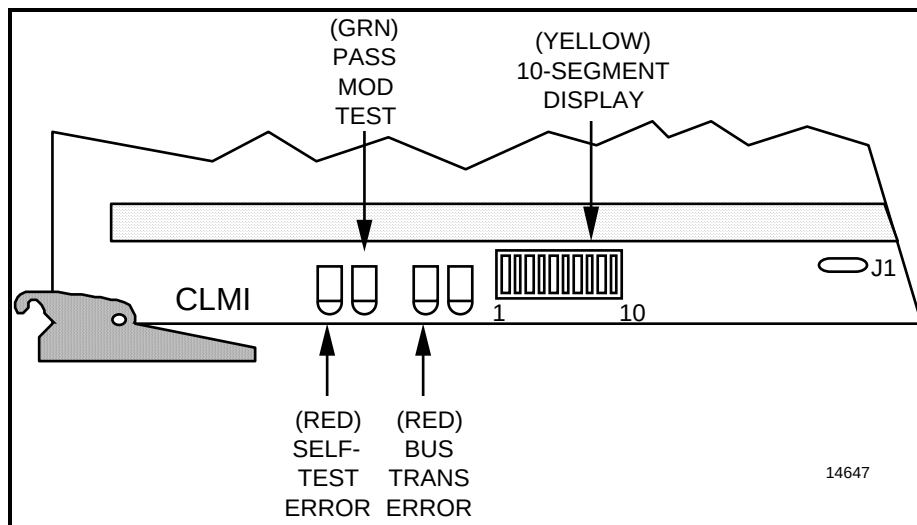


Figure 6.4.1-1 — CLMI Test and Status Indicators

Four LEDs are located on the front left edge of the CLMI board (see Figure 6.4.1-1). Each LED's definition and a brief description of its use follows. They are listed as shown in the figure, from left to right.

SELF-TEST or BOARD FAILURE (Red)

This LED lights under the following conditions:

- Module power coming on
- Module or CLMI reset sequence working
- CLMI self-test in progress
- Self-test failed
- Local parity errors

This LED is extinguished under the following conditions:

- Self-test is completed successfully
- Module or CLMI reset sequence completed successfully
- CLMI Abort Command sequence completed

PASSED MODULE TEST (Green)

This LED is lighted after the CLMI successfully completes the CPU tests.

BUS TRANSACTION ERROR (Red)

This LED lights under the following conditions:

- Module bus data parity error
- Bus error occurred during DMA access to module RAM

This LED is extinguished under the following conditions:

- Self-test is completed successfully
- Module power coming on
- Module or CLMI reset completed
- CLMI Abort Command in process

SOFTWARE RESET (Both LEDs OFF)

These LEDs are extinguished under the following conditions:

- The CLM platform software has issued a software reset of the board (that is, shutdown, failover)

10-SEGMENT DISPLAY (Yellow)

Also, on the front edge of the CLMI board, there is a 10-Segment display composed of yellow LEDs (see Figure 6.4.1-1). The function of each LED in this display, numbered 1 to 10 from left to right, are:

Segments 1, 2, and 3 monitor Field Port 1. Their functions are:

1. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
2. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
3. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.

Segments 4, 5, and 6 monitor Field Port 2. Their functions are:

4. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
5. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
6. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.

Segments 7, 8, and 9 monitor Field Port 3. Their functions are:

7. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
8. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
9. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.
10. **Not Used.**

6.4.2. CLMI Hardware Indicator Troubleshooting Chart

This chart may be used as an aid in determining the cause of a failure detected by the two fault indicators on the CLMI.

INDICATION	POSSIBLE CAUSE	SOLUTION
SELF-TEST ON after power-up or reset sequence	Missing CLMI I/O Paddleboard	Install CLMI I/O Paddleboard in the correct slot behind CLMI board.
	Failed CLMI or CLMI I/O Board	Substitute known good boards.
SELF-TEST goes OFF, then ON after the AM personality has been loaded	Incorrect Software Release	Check the Software Release.
TRANSACTION ERROR ON	Memory Board Error	Substitute known good memory board(s). If memory board(s) are OK, suspect the CLMI itself.

6.4.3. CLMI System Software Indications

System software indications that indicate hardware failures are presented in the Alarm Summary and Message Summary display, as well as in Custom schematics. A complete list of the error messages and meanings is supplied in the *Process Operations Manual* for the CLM.

6.4.4. CCP Hardware Indicators

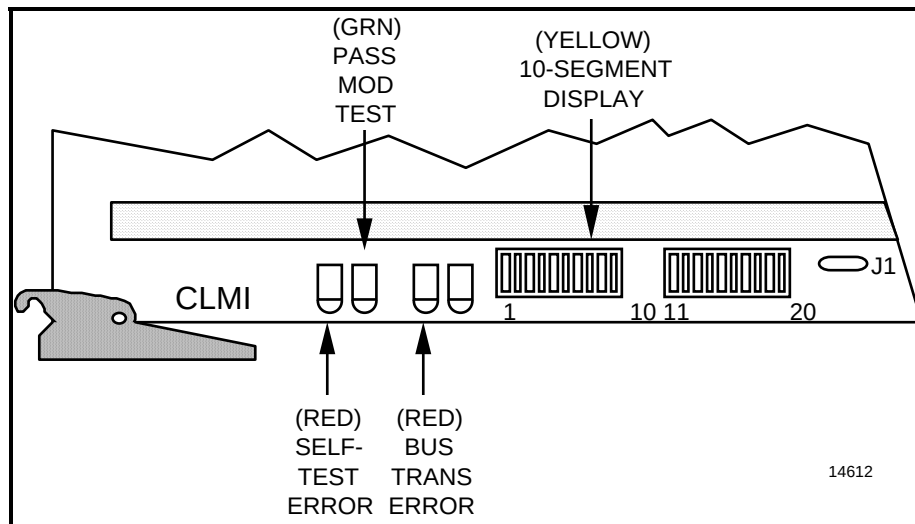


Figure 6.4.4-2 — CCP Test and Status Indicators

Four LEDs are located on the front left edge of the CCP board (see Figure 6.4.4-2). Each LED's definition and a brief description of its use follows. They are listed as shown in the figure, from left to right.

SELF-TEST or BOARD FAILURE (Red)

This LED lights under the following conditions:

- Module power coming on
- Module or CCP reset sequence working
- CCP self-test in progress
- Self-test failed
- Local parity errors

This LED is extinguished under the following conditions:

- Self-test is completed successfully
- Module or CCP reset sequence completed successfully
- CCP Abort Command sequence completed

PASSED MODULE TEST (Green)

This LED is lighted after the CCP successfully completes the CPU tests.

6.4.5. CCP Hardware Indicators

BUS TRANSACTION ERROR (Red)

This LED lights under the following conditions:

- Module bus data parity error
- Bus error occurred during DMA access to module RAM

This LED is extinguished under the following conditions:

- Self-test is completed successfully
- Module power coming on
- Module or CCP reset completed
- CCP Abort Command in process

SOFTWARE RESET (Both LEDs OFF)

These LEDs are extinguished under the following conditions:

- The CLM platform software has issued a software reset of the board (that is, shutdown, failover)

10-SEGMENT DISPLAY (Yellow)

Also, on the front edge of the CCP board, there is a 10-Segment display composed of yellow LEDs (see Figure 5-1a). The function of each LED in this display, numbered 1 to 20 from left to right, are:

Segments 1, 2, and 3 monitor Field Port 1. Their functions are:

1. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
2. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
3. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.

Segments 4, 5, and 6 monitor Field Port 2. Their functions are:

4. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
5. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
6. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.

Segments 7, 8, and 9 monitor Field Port 3. Their functions are:

- 7. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
- 8. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
- 9. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.

Segments 10, 11, and 12 monitor Field Port 4. Their functions are:

- 10. **TX RTS**—Illuminates when the transmit Request To Send is asserted.
- 11. **RX DATA**—Illuminates when Receive Data goes to a spacing condition.
- 12. **DSR/CD**—Illuminates when Data Set Ready and Carrier Detect are asserted.

Segments 13-20 are not used.

6.4.6. CCP Hardware Indicator Troubleshooting Chart

This chart may be used as an aid in determining the cause of a failure detected by the two fault indicators on the CCP.

INDICATION	POSSIBLE CAUSE	SOLUTION
SELF-TEST ON after power-up or reset sequence	Missing CCP I/O Paddleboard	Install CCP I/O Paddleboard in the correct slot behind CCP board.
	Failed CCP or CCP I/O Board	Substitute known good boards.
SELF-TEST goes OFF, then ON after the AM personality has been loaded	Incorrect Software Release	Check the Software Release.
TRANSACTION ERROR ON	Memory Board Error	Substitute known good memory board(s). If memory board(s) are OK, suspect the CCP itself.

6.4.7. CCP System Software Indications

System software indications that indicate hardware failures are presented in the Alarm Summary and Message Summary display, as well as in Custom schematics. A complete list of the error messages and meanings is supplied in the *Process Operations Manual* for the CLM.

6.5. SPARE PARTS

Spare parts for the Five Slot Module and its components (fan, power supply, boards, etc.) are listed in the *Five/Ten-Slot Module Service* manual in the *LCN Service - 1* binder.

Spare parts unique to the CLM are listed here.

PART NUMBER	DESCRIPTION	
*51196019-101	CLMI I/O ADAPTER	CLMI Paddleboard
*51305055-100	CLMI I/O ADAPTER	CLMI Paddleboard for RS-232/485
*51196877-100	CLMI-P	Communication Link Module Interface Board (256 Kbytes)
*51402426-200	CCP-1	Communications and Control Processor Board(1 mbytes)
*51402426-100	CCP-4	Communications and Control Processor Board(4 mbytes)
*51402436-100	Application dependent	Universal Communication Interface Module
*51107403-100	LCN I/O	LCN Adapter Board Assembly
*51400667-100	LCNI	Local Control Network Board
51401291-100	LLCN	Low Power Local Control Network Board
*51400910-100	EMEM	Enhanced Memory Board, 1 M-word
*50401072-200	QMEM-2	Memory board 2 M-Word
*50401072-300	QMEM-3	Memory board 3 M-Word
*50401072-400	QMEM-4	Memory board 4 M-Word
51401288-100	HPK2-2	High Performance Kernel Processor/2 Mw Mem.
*51401288-200	HPK2-3	High Performance Kernel Processor/3 Mw Mem.
*50401551-400	K2LCN-4	K2LCN Processor board/ 4 Mw mem
*50401551-600	K2LCN-6	K2LCN Processor board/ 6 Mw mem
*50401551-800	K2LCN-8	K2LCN Processor board/ 8 Mw mem

* ORU Level Replacement Item

NOTE

Cable lengths and appropriate pinning for RS-232 Data Cables are listed in Appendix A.

CLM CONNECTIONS TO BRISTOL BABCOCK RTUs

Appendix A

A.1 BEFORE YOU START

Installation of a Bristol-Babcock Network 3000 requires the following Bristol Babcock publications or their equivalents:

No. CI-3330 for the Distributed Process Controller - 3330

No. CI-3335 for the Distributed Process Controller - 3335 and 3310

If modem and/or interface converters are used, you will also need their hardware installation manuals. The CLM to Bristol Babcock Network 3000 connection has been tested with the following network components:

- Direct connect from CLM to DPC-3330, 3310, and 3335.
- Direct connect from CLM to PC, 9, and 25 pin connectors.
- UDS 202S/T modem to UDS 202S/T modem, 4-wire private line.
- UDS 202S/T modem to Bristol Babcock Private Line Modem.
- Black Box Corp. RS-232<-->RS-422/485 Interface Converter (product number IC007A).
- Black Box Corp. Telco Line Bridge (TLB) (product number TL205).
Single level - CLM to Modem to TLB to Modem to RTU
Two levels - CLM to Modem to TLB to TLB to Modem to RTU

Future testing will include radio and fiber optic modems.

A.2 BRISTOL BABCOCK RTU ADDRESSING

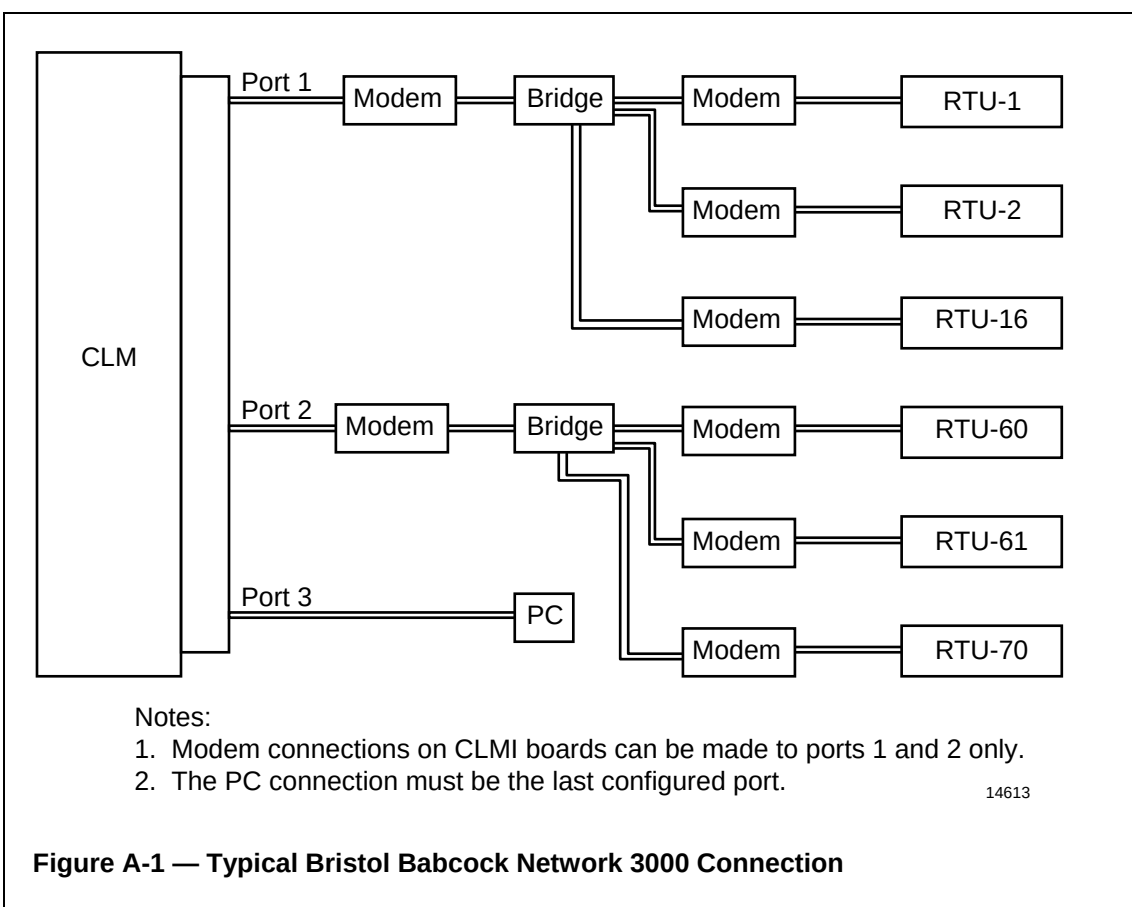
Each CLMI board can be connected to one Bristol Babcock Network 3000. This connection can consist of 1, 2, or 3 ports connected to different branches of the network and one port connected to a PC for support of the Bristol Babcock Toolkit functions. Modem control is provided on CLM ports 1 and 2 for CLMI and all 4 ports for CCP. Therefore, on a CLMI, ports 3 and 4, if used, must be direct connects. The PC must be connected to the last port configured on the CLMI I/O adapter or paddleboard. This is the highest numbered port used.

Because all port connections are treated as one Bristol Babcock Network, the Remote Terminal Unit (RTU) addresses must be configured as one network. This means that RTU addresses must be consecutive on a port and consecutive across ports. However, spare addresses can, and should, be reserved on each port.

The CLM currently supports flat network both flat and multi-level networks. However, the CLM only communicates to the first level and relies on the ACCOL language and configuration in the RTU to communicate to the lower level RTUs.

Important: The CLM does not support a master-based RTU. In all cases, the CLM must be the master.

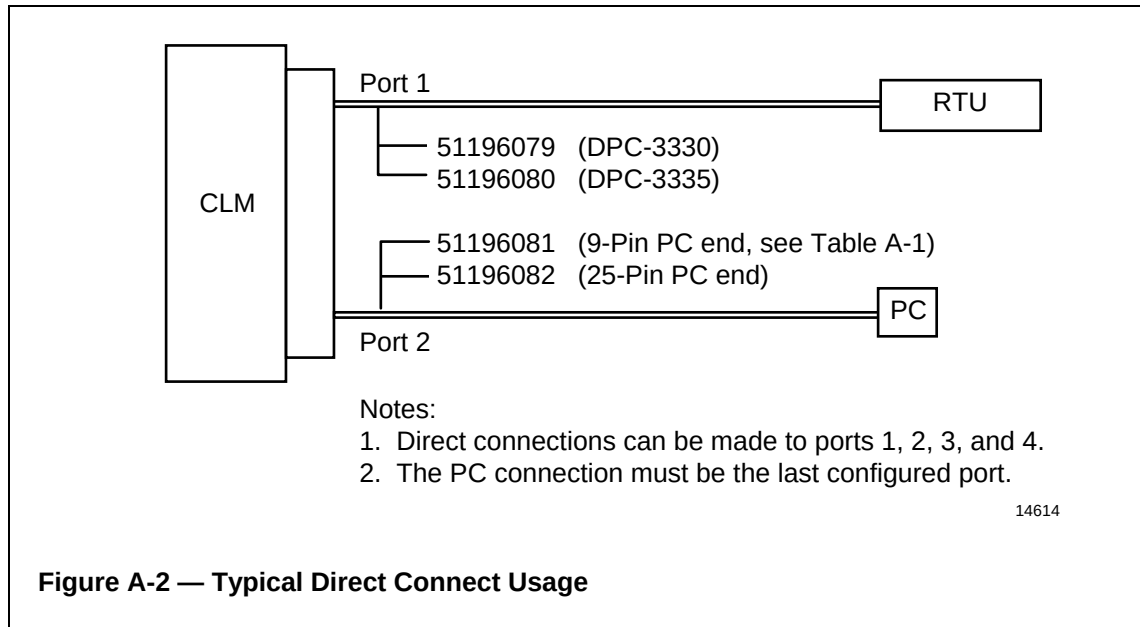
Figure A-1 illustrates examples of connecting a nonredundant CLM to a Bristol Babcock Network 3000. In this example RTU addresses 1–59 are reserved for Port 1. On Port 2, RTU addressing starts at 60 and can extend to 127. These addresses are switch settings (SW2) on the DPC-3330/3335 CPU Engine Board of the RTU.

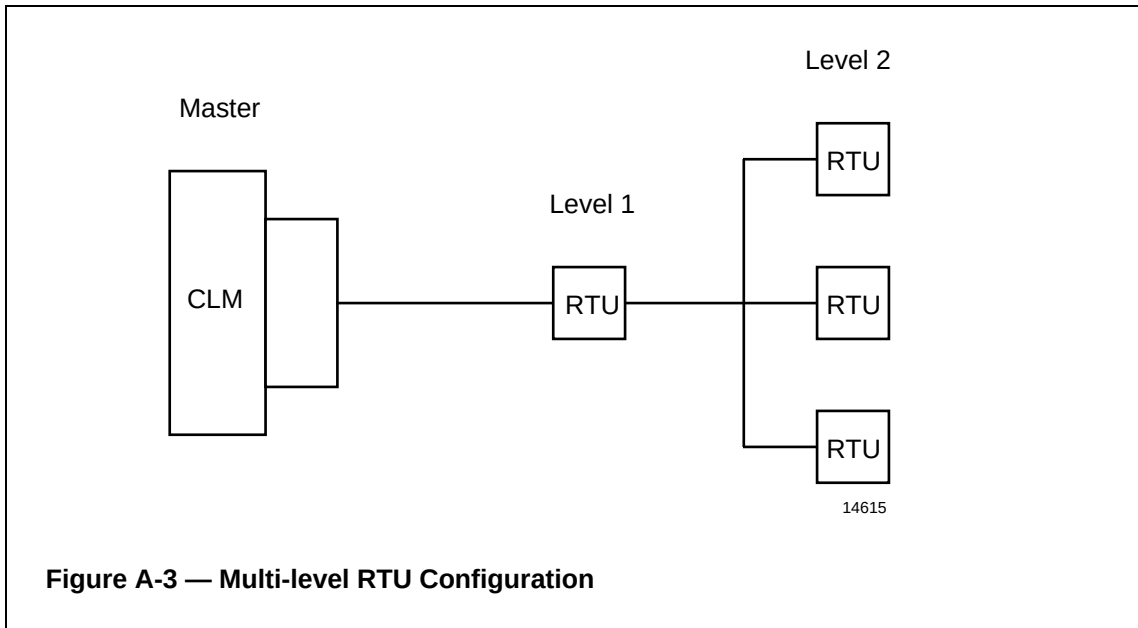


A.3 CLM TO BRISTOL BABCOCK RTU CONNECTIONS

The CLM can be either a direct-connection or connected by way of modems, bridges, and interface converters to the Bristol Babcock Network 3000.

Figure A-2 illustrates examples of directly connecting a nonredundant CLM to a Bristol Babcock Network 3000.





Figures A-4 through A-7 illustrate the cables for directly connecting a CLM to a Bristol Babcock Network 3000.

Figure A-4 illustrates the cables for directly connecting a CLM to a Bristol Babcock DPC-3330.

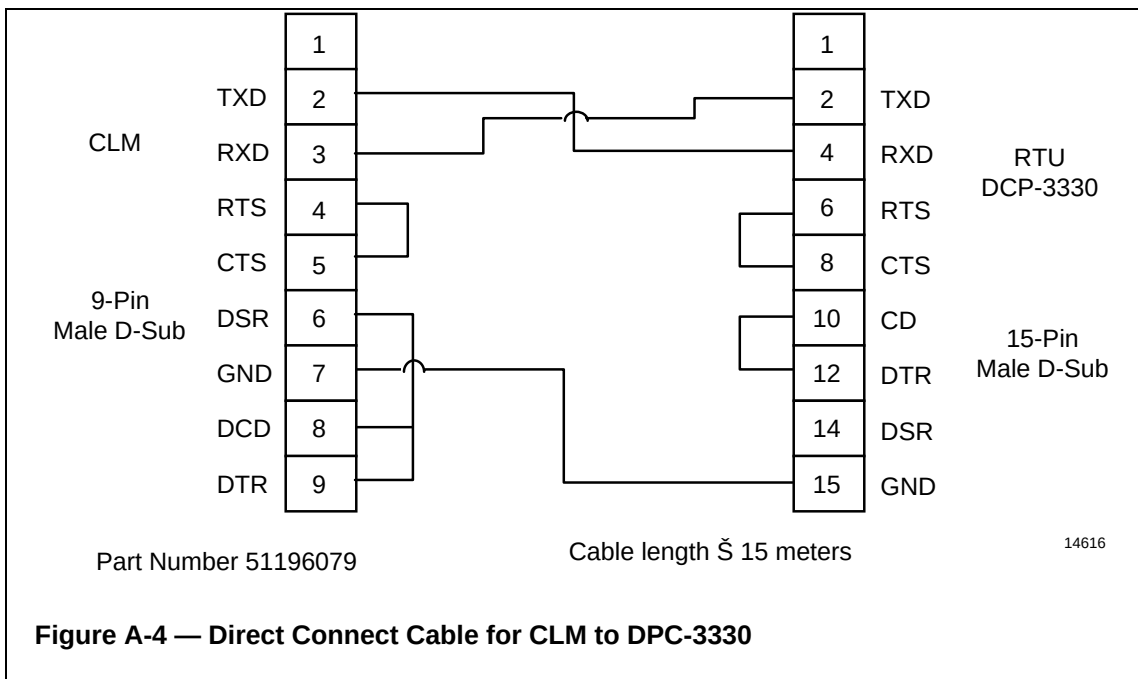


Figure A-5 illustrates the cable for directly connecting a CLM to a Bristol Babcock DPC-3335.

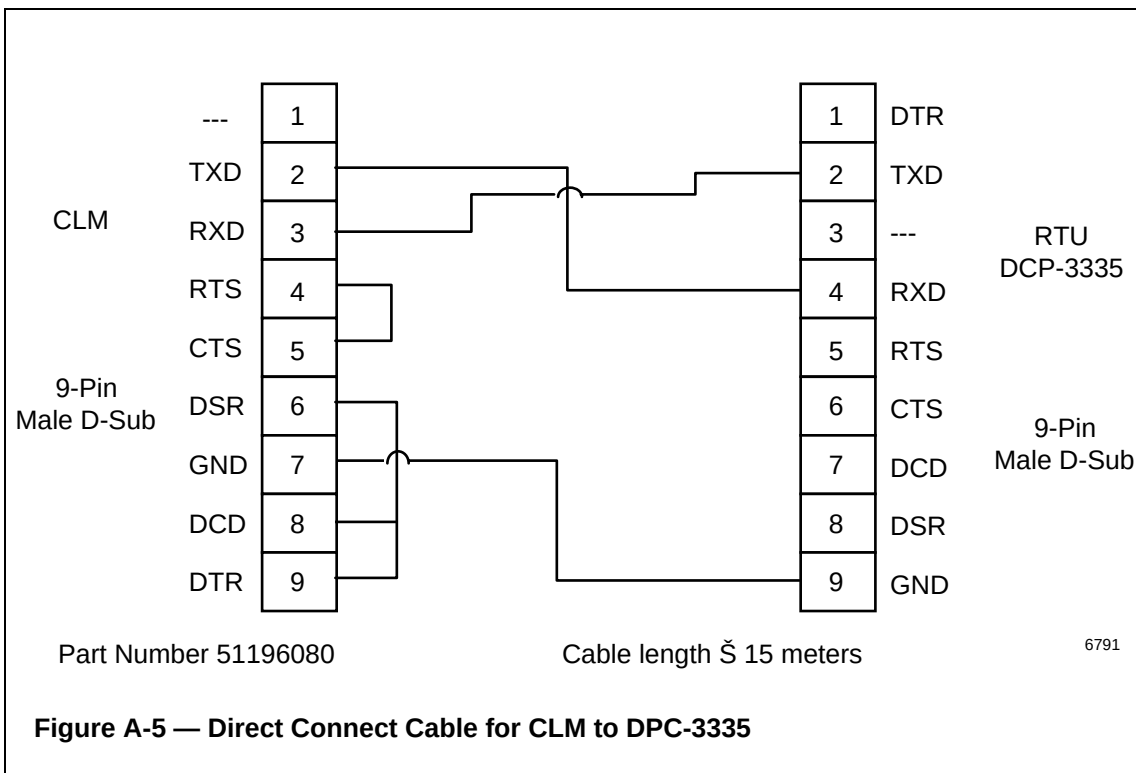


Figure A-6 illustrates the cable for directly connecting a CLM to a 9-pin PC connector to support Bristol Babcock Toolkit functions.

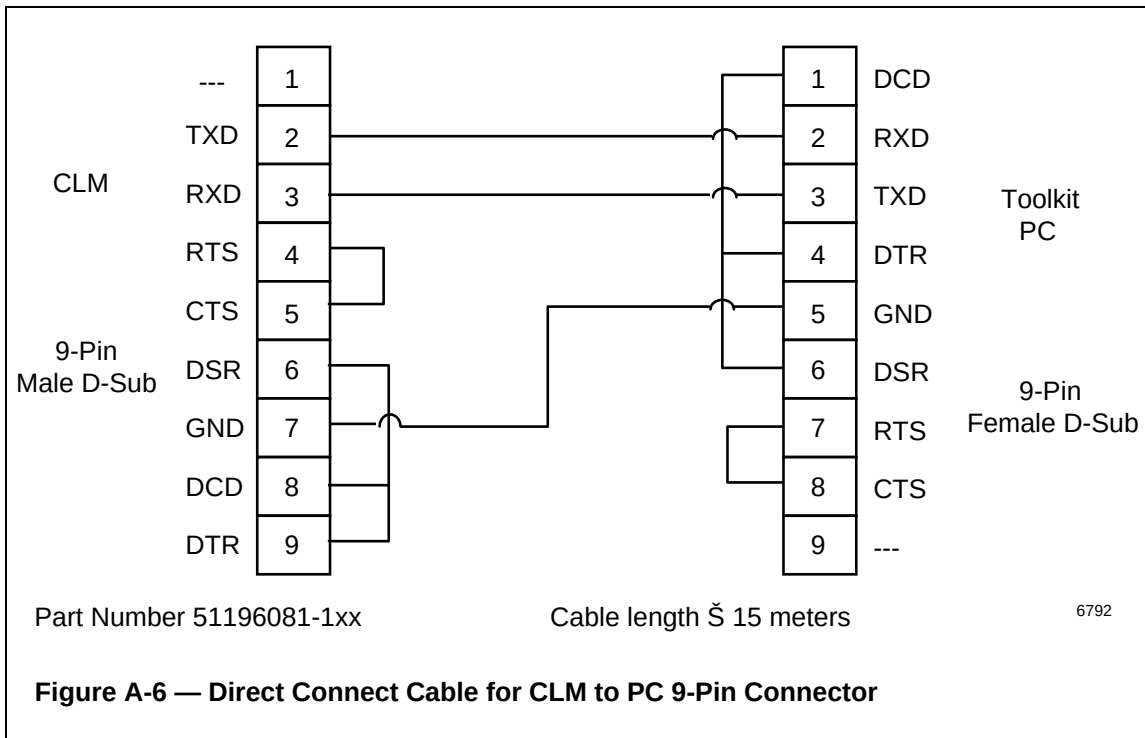


Figure A-7 illustrates the cable for directly connecting a CLM to a 25-pin PC connector to support Bristol Babcock Toolkit functions.

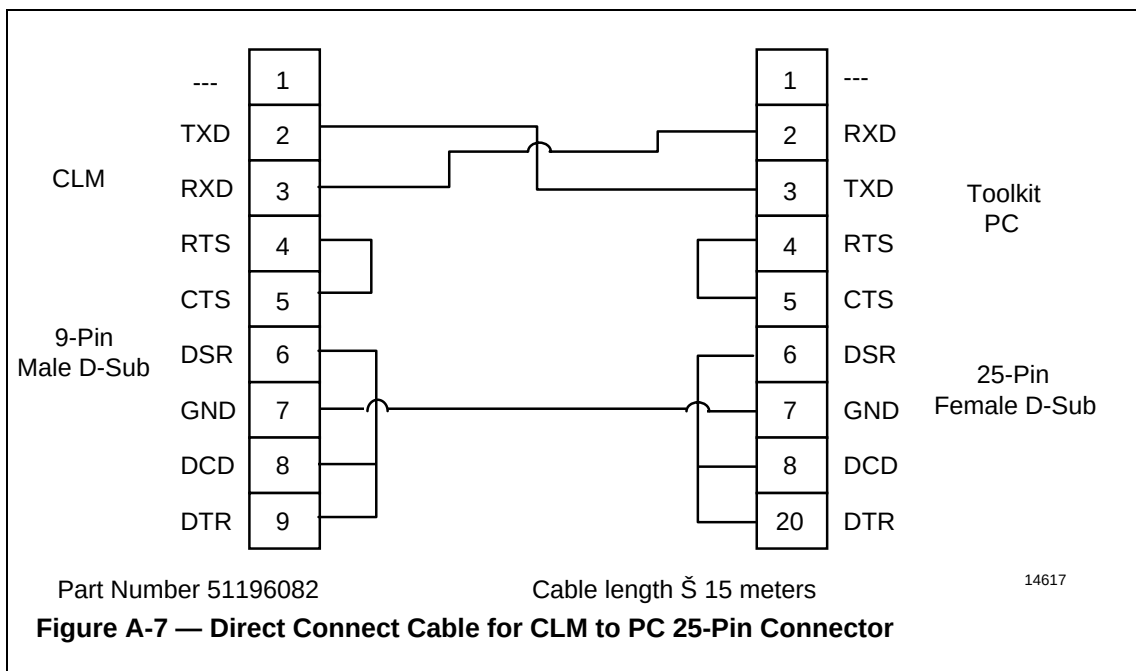


Figure A-8 illustrates examples of connecting a nonredundant CLM to a Bristol Babcock Network 3000 through modems.

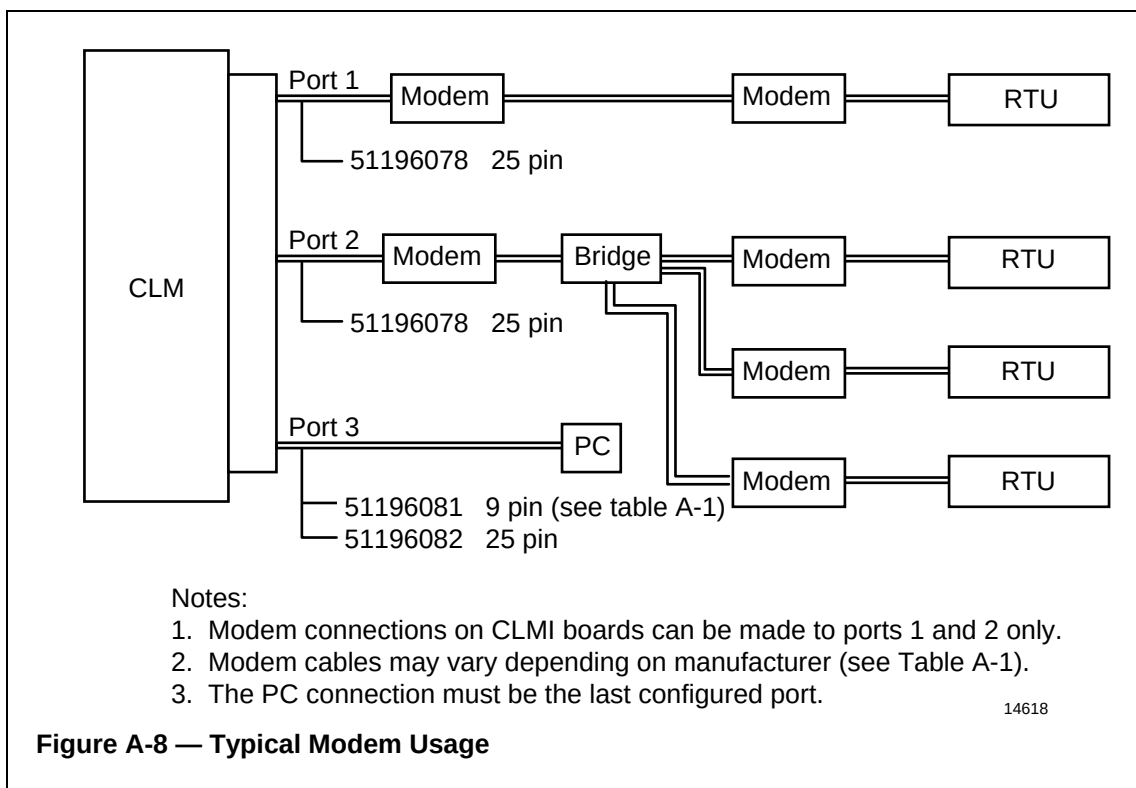


Figure A-9 illustrates examples of connecting a nonredundant CLM to a Bristol Babcock Network 3000 through an RS-232 to RS-485 converter.

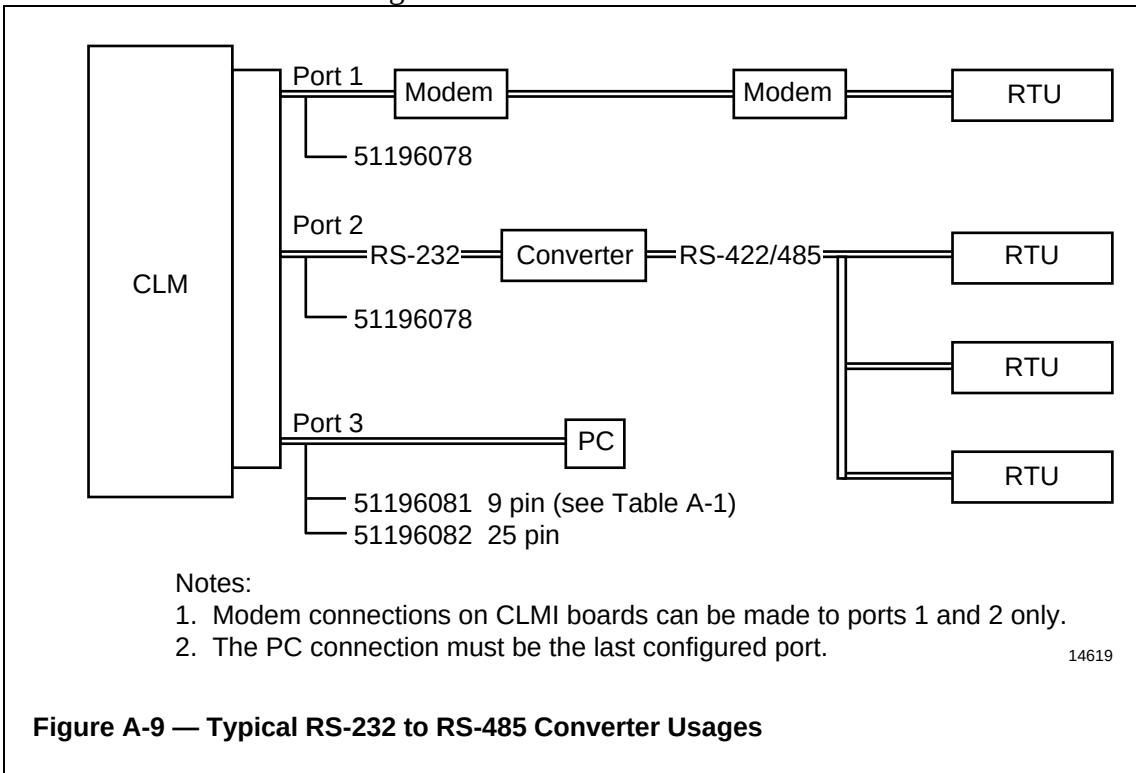


Figure A-10 illustrates the modem cable for connecting a CLM to a 25-pin modem or interface converter for a Bristol Babcock Network 3000 connection.

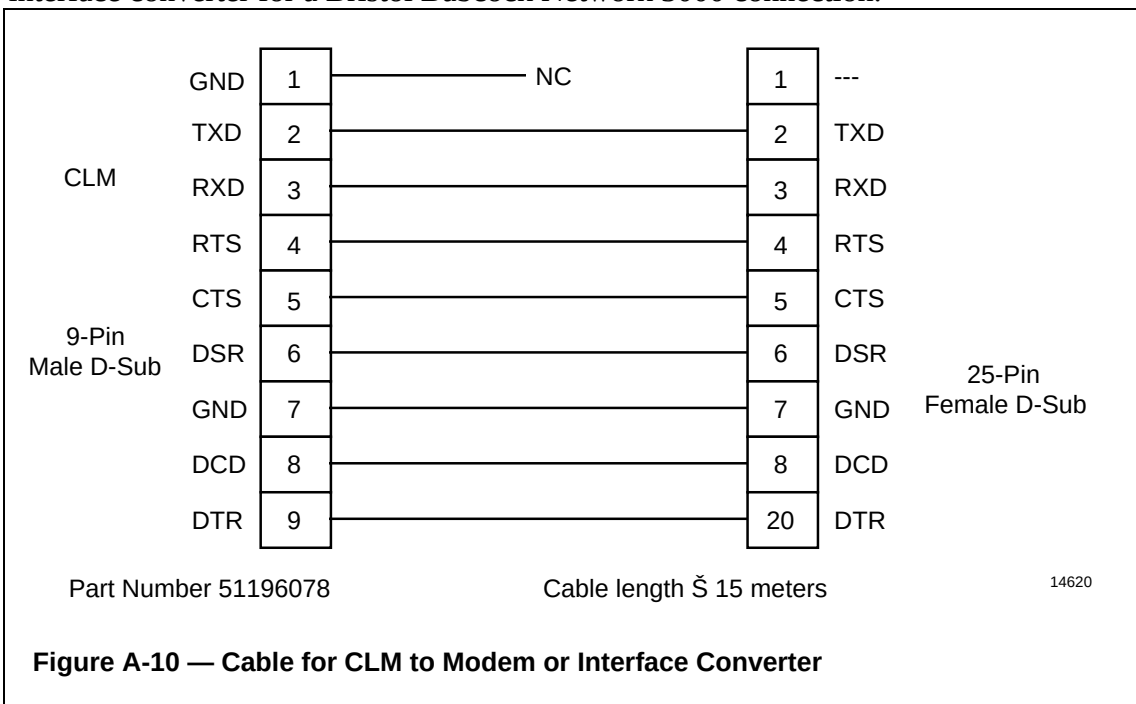
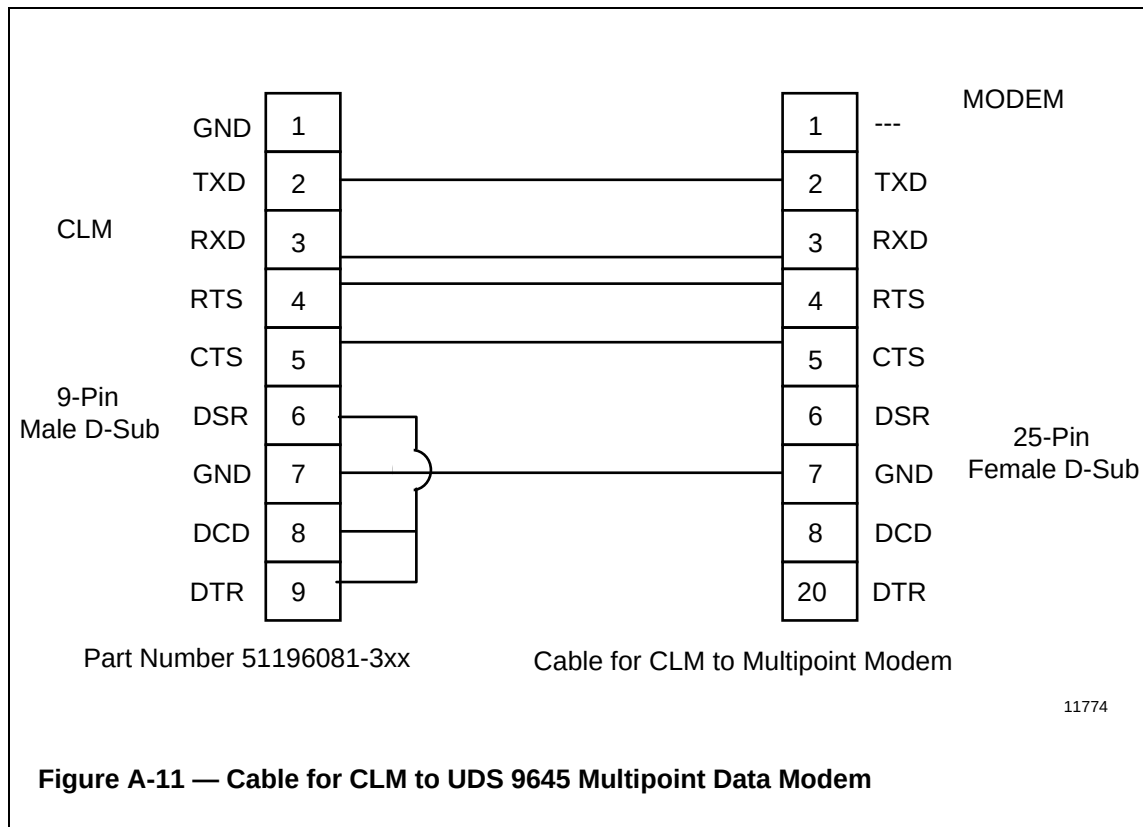


Figure A-11 illustrates the cable for connecting a CLM to a multipoint modem.



A.4 STANDARD CABLES for CLM to BRISTOL BABCOCK NETWORK 3000 CONNECTIONS

Table A-1 lists the standard cables used in connecting the CLM to a Bristol Babcock Network 3000 and direct connect cables for PC and DPC-3330/3335 connections.

Table A-1 — Standard Cables

Part Number	Description
51196078-xxx	CLM to 25-pin male modem end
51196079-xxx	CLM direct connect to 15-pin male DPC-3330 end
51196080-xxx	CLM direct connect to 9-pin male DPC-3335 end
51196081-1xx	CLM direct connect to 9-pin female PC end
51196081-2xx	CLM direct connect to 9-pin male PC end
51196081-3xx	CLM to Multipoint Modem
51196082-xxx	CLM direct connect to 25-pin female PC end
xxx is the cable length in meters. Maximum length is 15 meters. Standard xxx codes and cable lengths are: 03 - 3 meters 05 - 5 meters 15 - 15 meters	

CLM CONNECTIONS TO MODBUS AND MODBUS COMPATIBLE DEVICE NETWORKS

Appendix B

B.1 BEFORE YOU START

You will need the hardware installation manuals for the Modbus or Modbus compatible device or device network that you are connecting to the CLM. If modem and/or interface converters are used in the interface connection, you will also need their hardware installation manuals.

B.2 CLMI and CCP PORT CONNECTIONS

Each CLMI board provides for up to four interface port connections. Each port can be connected to a different device or network of devices. The CLMI board only provides modem control on ports 1 and 2. Therefore, ports 3 and 4 can only be used for direct device or network connections.

Each CCP board provides for up to four interface port connections. Each port can be connected to a different device or network of devices. The CCP board provides modem control on all four ports.

B.3 CLM TO MODBUS COMPATIBLE CONNECTIONS

The CLM can be either a direct connection or connected by way of modems, bridges, and interface converters to the Modbus compatible device or network.

Table B-1 lists standard cables used in connecting the CLM to Modbus compatible devices. Cables may vary depending on modem and device manufacture.

Table B-1 Standard Cables

Part Number	Description
51196078-xxx 51196082-xxx 51196081-3xx	CLM to 25-pin male modem end CLM direct connect to 25-pin female end CLM to Multipoint Modem
xxx is the cable length in meters. Maximum length is 15 meters. Standard xxx codes and cable lengths are: 03 - 3 meters 05 - 5 meters 15 - 15 meters (maximum length)	

Figure B-1 illustrates examples of connecting a nonredundant CLM to Modbus compatible devices.

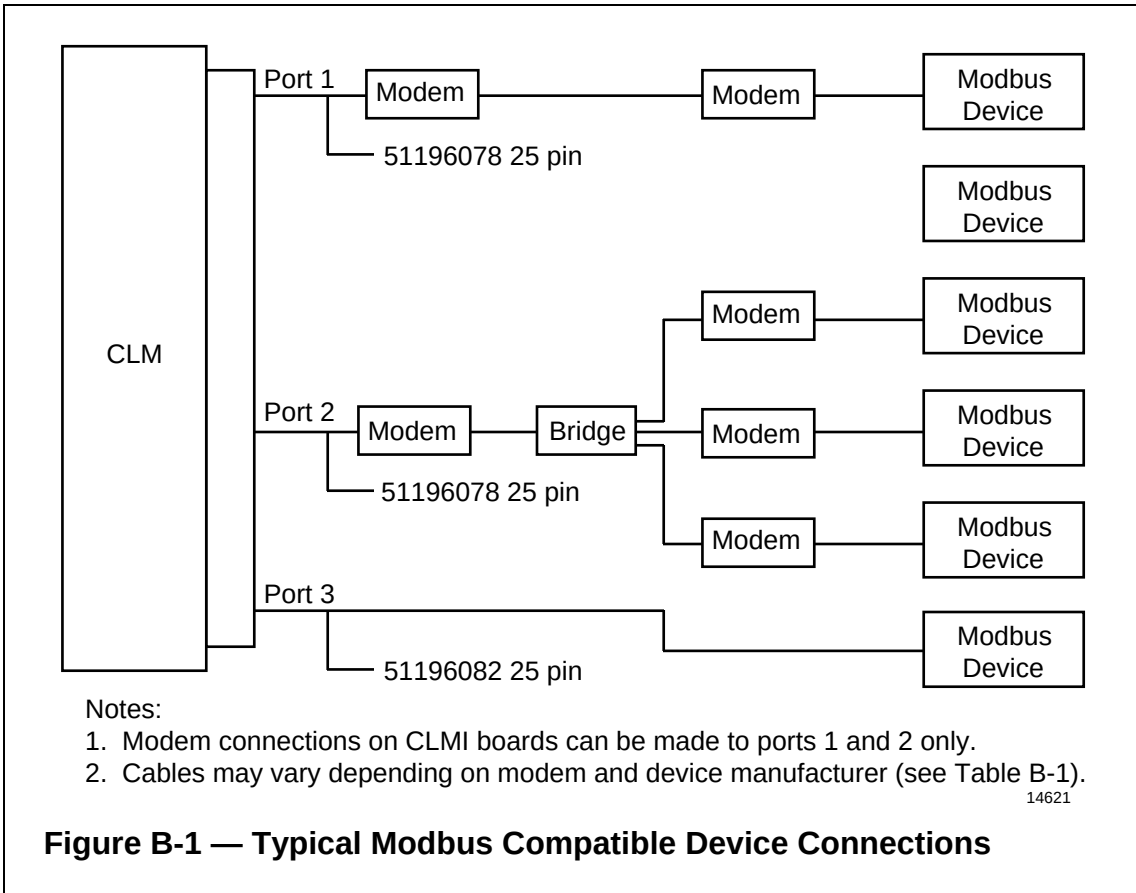


Figure B-2 illustrates a typical cable for directly connecting a CLM to a 25-pin Modbus compatible device. The illustrated cable is the standard cable part number 51196082.

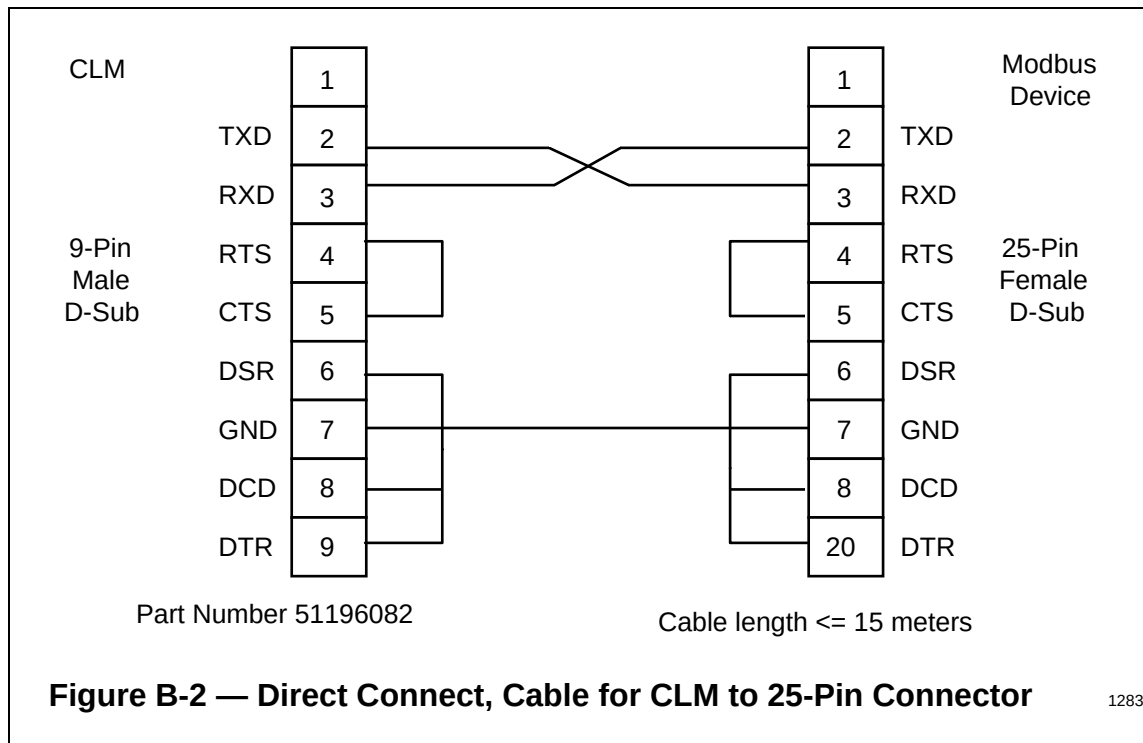


Figure B-3 illustrates the modem cable for connecting a CLM to a 25-pin modem or interface converter.

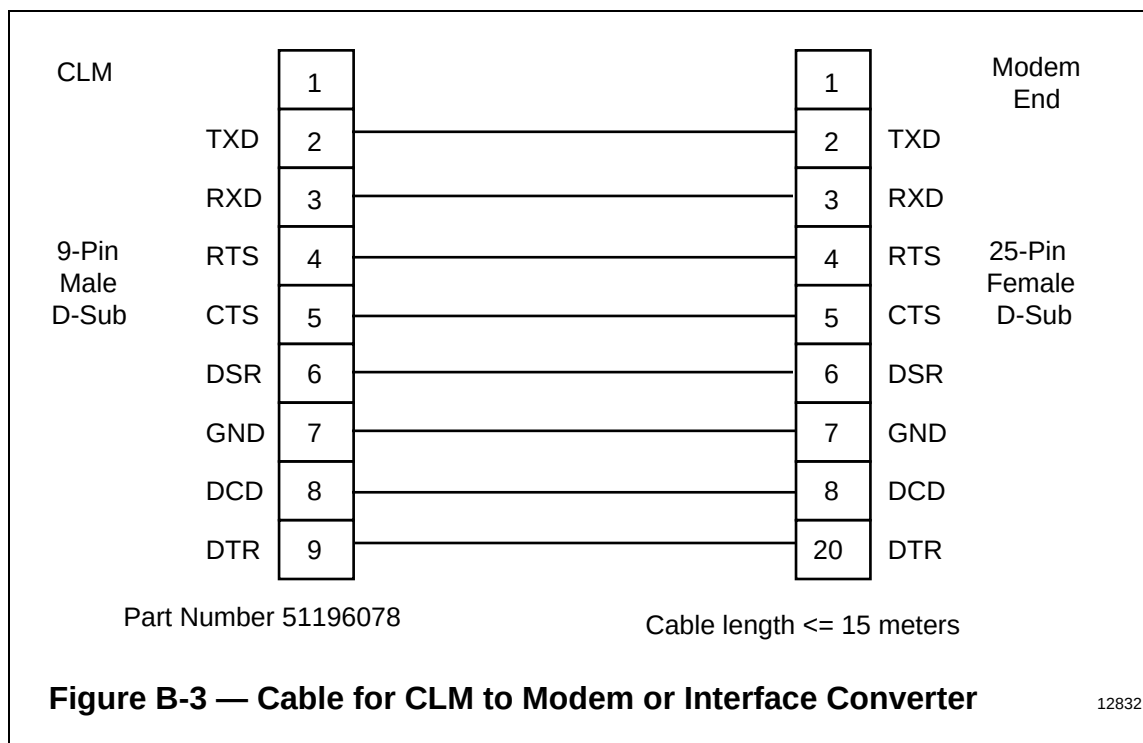
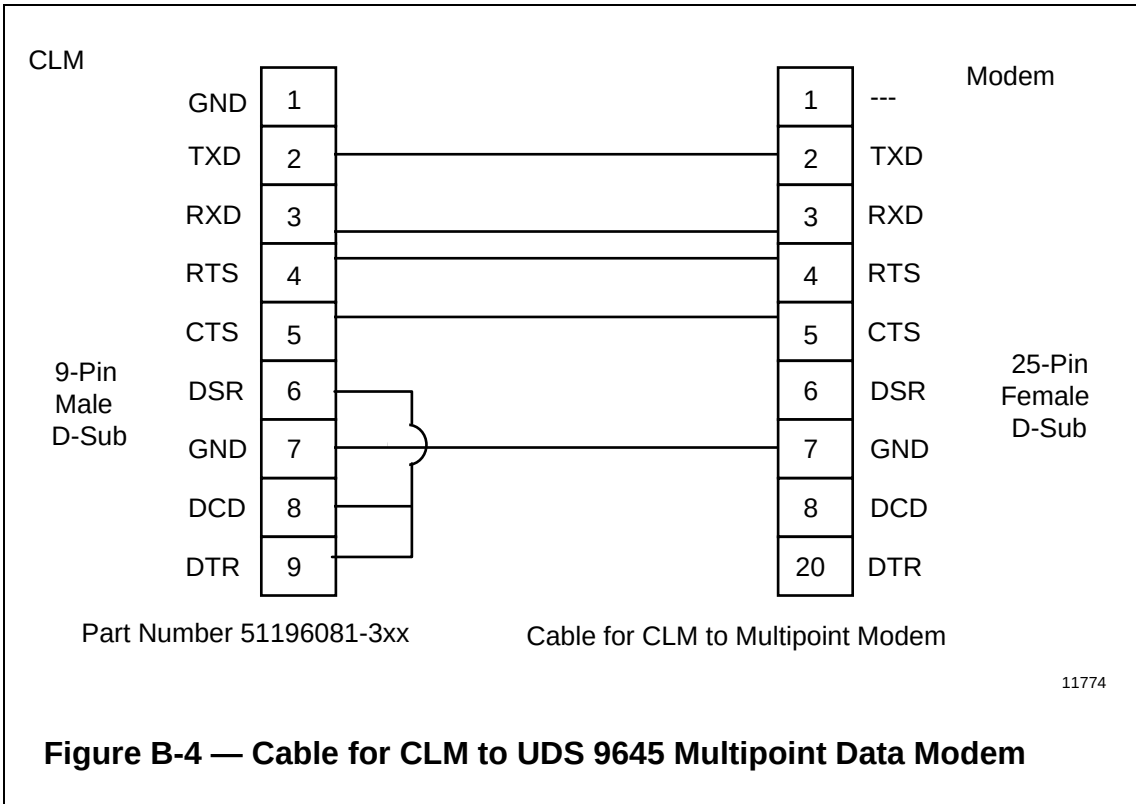


Figure B-4 illustrates the cable for connecting a CLM to a multipoint modem.



UCLM: CCP AND UCIM REDESIGNED

Appendix C

CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

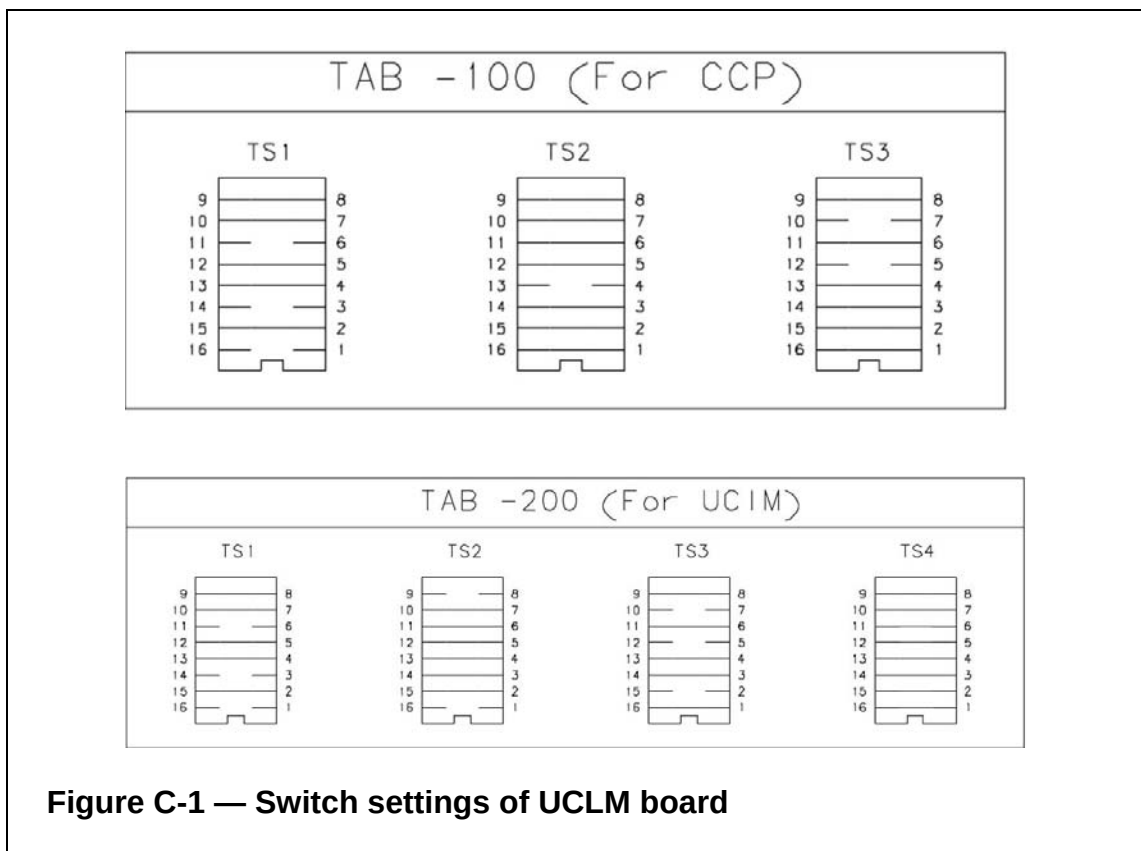
C.1 ORDERING INFORMATION

To order the boards use the Honeywell part numbers as provided in the following table:

Part Number	Board Name
51306409-100	CCP
51306409-200	UCIM

C.2 SWITCH SETTINGS OF UCLM BOARD

Figure C-1 illustrates the redesigned CCP and UCIM settings.



There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM.

Index

1

10-Segment display, 42, 45

A

AC voltage options, 9

AM processor board and the .ii.CLMI or CCP board, 40

Application dependent Universal Communication Interface Module, 47

Application Module Specification and Technical Data, 7

assembling, 13

assembly, 13

B

board complements, 38

BOARD FAILURE, 41, 44

Board Failure LED, 41, 44

Broadband modems, 11

BUILDER, 36

BUS TRANSACTION ERROR (RED), 42

BUS TRANSACTION ERROR (Red), 45

C

cable lengths, 56

cable lengths, 57

Cable lengths, 10

cabling, 13

Cabling restrictions/considerations, 10

Carrier Detect, 42, 45

CCP board, 44

CCP-P, 47

chassis-ground, 13

CLM Parameter Reference Dictionary, 7

CLM Planning, Installation, and Service (this publication), 7

CLM Platform Implementation, 7

CLM Process Operations, 7

CLM spare parts, 47

CLMI, 47

CLMI and CCP Troubleshooting, 41

CLMI board, 41, 49

CLMI I/O ADAPTER, 14, 47

CLMI I/O ADAPTER CLMI Paddleboard for RS-232/485, 47

CLMI-P Communication Link Module Interface Board, 47

Configuration overview, 21

Configurations, 10

connecting a CLM to a 25-pin modem, 55

connecting a CLM to a 25-pin modem or interface converter., 59

connecting a CLM to a 25-pin PC connector, 54

connecting a CLM to a 9-pin PC connector, 53

connecting a CLM to a Bristol Babcock DPC-3330, 52

connecting a CLM to a Bristol Babcock DPC-3335, 53

connecting a CLM to a Bristol Babcock Network 3000, 51

connecting a CLM to a multipoint modem, 60

connecting a nonredundant CLM to a Bristol Babcock Network 3000, 50, 51, 55

connecting a nonredundant CLM to a Bristol Babcock Network 3000 through modems, 54

connecting a nonredundant CLM to Modbus compatible devices, 58

connection to a modem, 17

Core Module Test System, 7

D

Data Set Ready, 42, 45

direct connect cables, 56

Direct connection, 11

directly connecting a CLM to a 25-pin Modbus compatible device, 59

DSR/CD, 42, 45

DSR/CD—Illuminates when Data Set Ready and Carrier Detect are asserted., 45

E

Electrostatic Discharge (ESD) procedures, 40

EMEM, 47

environmental constraints, 8

F

failed board, 40

failure detected by the two fault indicators, 43, 46

fan assembly, 40

Field adjustments, 40

Field Services, 6

Five/Ten-Slot Module Service manual, 20, 40

Five/Ten-Slot Module Service manual, 38

Five/Ten-Slot Module Service Manual, 7

five-slot or dual node module, 38

flat network, 50

Frequency, 9

front cover of the CLM module, 20
functional circuit board, an .ii.input/output card and
special cabling have been added to this standard
module., 38

G

green LEDs, 20
grounding, 13

H

Hardware Verification Test System, 7
HPK2-2, 47
HPK2-3, 47

I

inrush current, 9
install the CLM module, 13
Installation Support, 6
Installation wrap-up, 19

K

K2LCN, 15
K2LCN-4, 47
K2LCN-6, 47
K2LCN-8 K2LCN Processor board/ 8 Mw mem,
47

L

LCN I/O, 47
LCN I/O paddleboard, 14
LCN Site Planning, 7
LCN Site Planning manual, 6
LCN System Checkout, 7
LCN System Installation, 7
LCN System Installation manual, 6
LCNI, 47
LLCN, 47
logic-ground, 13

M

Maintenance Test Operations, 7
Modem selection, 11
modem usage, 12
Modem usage, 11
Modem, wiring, 17
Modems, pinning, 17
modems, telephone, 11
module installation, 13

N

NCF update process, 32
network components, 49
node address, 14
nonredundant CLM port connections, 12

O

Optimum Replaceable Unit, 40
ORU, 40
*overview of the CLM equipment and lists
reference documents available from
Honeywell.*, 6

P

parity jumper, 14
Passed Module Test LED (green), 42, 44
planning and installation, 6
port cabling, 17
port cabling configurations, 10
port connections, 12
power supply, 40
Power-on tests, 20
preliminary checks, 40
Process Operations Manual, 7

R

Receive Data, 42, 45
red LEDs, 20
Redundant CLM Restrictions/Rules, 39
Related publications, 7
repair, 38
Request To Send, 42, 45
Required publications, 49
RS-232 cable, 11
RS-232 to RS-485 converter, 55
RS232/485, 16
RTS/CTS delay, 18
RX DATA, 42, 45
**RX DATA—Illuminates when Receive Data
goes to a spacing condition.**, 45

S

Segments 1, 2, and 3 monitor Field Port 1, 42, 45
Segments 10, 11, and 12 monitor Field Port 4, 45
Segments 4, 5, and 6 monitor Field Port 2, 42, 45
Segments 7, 8, and 9 monitor Field Port 3, 42, 45
SELF-TEST, 41, 43, 44, 46
Self-test LED, 41, 44
service instructions, 38
short-haul modem, 11

Site requirements, 8
SOFTWARE RESET, 42, 45
Spare parts, 47
Spare parts, CLM table of, 47
standard cables, 56
standard cables, 57
Storage conditions, 8

T

Technical Support, 6
telephone modems, 11
test, 38
Test System Executive, 7
Tests, 20
toll-free number, 6
Troubleshooting, general, 40
TX RTS, 42, 45
**TX RTS— Illuminates when the transmit
Request To Send is asserted.**, 45

U

unpacking, 13
updating the LCN Network Configuration File, 32

V

vibration, 8

W

Warning on ESD procedures, 40

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